

Hugging Face Model Research and Acquisition Guide

Authoritative, implementation-ready research for the four-core detection architecture

Research verified on	15 August 2026
Document version	1.0
Hub repositories investigated	29
Supplied PDF pages analyzed	156
Scope	Research and acquisition only - no product model implementation

Research first. Verify second. Recommend third. Implementation comes later.

Prepared from the executable repository, six supplied architecture PDFs, official Hugging Face Hub repositories and authoritative upstream sources.

2. Document control and evidence standard

Title	VeriTrust Hugging Face Model Research and Acquisition Guide
Version	1.0
Research date	15 August 2026
Workspace	C:\Users\progr\OneDrive\Desktop\Projects\VeriTrust-Site-main
Decision authority	Executable code = current state; handbooks = target intent only where explicitly future-facing
Hub evidence	Official model API, model card, immutable revision, file list, config/tokenizer/processor files, provider mapping
Uncertainty labels	UNKNOWN, UNVERIFIED, REQUIRES BENCHMARKING, REQUIRES LEGAL REVIEW, NOT IMPLEMENTATION READY
Production status	No public model is designated production-ready in this report

All availability/provider statements are point-in-time facts as of the verification date. Hub repositories, provider mappings, files and licenses can change. Acquisition must use the recorded immutable revision rather than a mutable branch.

Evidence interpretation

- Published/card metrics are reported as claims from the repository owner, not VeriTrust qualification evidence.
- Exact Hub file sizes are used where present. Parameter-memory figures are mathematical lower bounds only; runtime RAM/VRAM is higher.
- A model license and its training-dataset license are separate. Every commercial-use statement remains subject to legal review.
- Different repository names do not prove independent evidence. Architecture, tokenizer and dataset overlap are evaluated explicitly.
- No model ID should enter a current environment variable unless both transport and output semantics match the current adapter.

3. Executive summary

VeriTrust does not need one generic AI model. It needs a portfolio of narrowly defined evidence producers, plus deterministic preprocessing, validation, quality gates, calibration and versioned gateway correlation. The current repository is a configurable proof-stage runtime with three live provider shapes and several scientifically unsafe normalization/fallback shortcuts. The target handbooks require a materially different registry- and evidence-led architecture.

The most important acquisition finding is that the handbook seeds mostly exist, but almost none are safe drop-ins for today's adapters. The only investigated task model with a currently live HF image-classification mapping and native Real/Fake labels is a concept-drifted legacy ViT useful as a compatibility fixture. The live MailGuard-compatible seed is a mixed email/URL four-class DistilBERT, not a clean binary email model. Neither recommended URLBERT model currently has an HF provider mapping, and the current Swift label map reverses defacement and phishing.

Question	Answer
What exists on HF?	29 repositories were opened and inspected; 21 justify controlled evaluation and 8 are rejected.
What works today?	Only a small compatibility subset can use today's generic serverless transport; scientific role, label and calibration gaps still prevent production activation.
What best implements target intent?	Dedicated preprocessing/model workers, immutable registry resolution before inference, task-specific evidence contracts, private structured models, calibration and abstention.
Gateway model?	No learned gateway classifier initially. Use deterministic, versioned correlation rules.
Learning Module model?	No core Hugging Face model. It is course/content/assessment/certification infrastructure.

Immediate research stack: YuNet preprocessing; yermandy facial-manipulation detector; official Community Forensics 384 synthetic-image detector; Sonje03 fast phishing classifier; mDeBERTa-v3 multilingual fine-tune; ModernBERT shadow; URLBERT-v4 multiclass; pirocheto ONNX lexical shadow; DeBERTa-v3-small page-text fine-tune; SigLIP2 approved-brand embeddings. Independently train VeriTrust forensic, OOD and structured-feature models.

4. Get these models first - one-page acquisition sheet

Priority	VeriTrust role	Exact repository/component	Exact URL	Params	Weight/repo size	Class	Action
P0	Synthetic full-frame	OwensLab/commfor-model-384	https://huggingface.co/OwensLab/commfor-model-384	21.8M	83.22 MiB repo	CLASS A	DOWNLOAD + SELF-HOST
P0	Facial manipulation	yermandy/deepfake-detection	https://huggingface.co/yermandy/deepfake-detection	UNKNOWN detector; CLIP-L backbone	1.70 GiB repo	CLASS A	DOWNLOAD + SELF-HOST
P0	Fast phishing text	Sonje03/phishlens-distilbert	https://huggingface.co/Sonje03/phishlens-distilbert	67.0M	255.43 MiB	CLASS A	DOWNLOAD + SELF-HOST
P0	Multiclass URL	CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	https://huggingface.co/CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	3.69M	14.08 MiB	CLASS A	DOWNLOAD + SELF-HOST
P1	Multilingual phishing	microsoft/mdeberta-v3-base	https://huggingface.co/microsoft/mdeberta-v3-base	~279M	5.45 GiB repo	CLASS B	FINE-TUNE FIRST
P1	Phishing shadow	mikaelnurminen/modernbert-large-phishing	https://huggingface.co/mikaelnurminen/modernbert-large-phishing	395.8M	4.20 GiB repo	CLASS A	BENCHMARK SHADOW
P1	Independent URL shadow	pirocheto/phishing-url-detection	https://huggingface.co/pirocheto/phishing-url-detection	LinearSVM	1.49 GiB repo	CLASS A	DOWNLOAD ONNX ONLY
P1	Page-text backbone	microsoft/deberta-v3-small	https://huggingface.co/microsoft/deberta-v3-small	~142M	3.59 GiB repo	CLASS B	FINE-TUNE FIRST
P1	Brand screenshot embedding	google/siglip2-base-patch16-224	https://huggingface.co/google/siglip2-base-patch16-224	375.2M	2.83 GiB repo	CLASS A	DOWNLOAD + SELF-HOST
P1	Face boxes/landmarks	OpenCV Zoo YuNet (non-HF)	-	~0.1M class	small ONNX	CLASS D	PIN NON-HF COMPONENT
P1	Forensic + OOD + structured	Private VeriTrust artifacts	-	Role-specific	TBD	CLASS C	TRAIN PRIVATE MODELS

DO NOT configure these IDs in production environment variables yet. The target shortlist requires adapter, label-map, registry, benchmark and calibration work described in the handoff section.

5. Repository audit scope and verification

The workspace was recursively inventoried. First-party text/code was semantically reviewed, with line-level review concentrated on every runtime path that constructs, sends, normalizes, persists, displays or documents model inference. Vendored packages were inventoried by declared/locked version and exercised usage, not treated as first-party architecture.

Audit measure	Observed
Directories inventoried	87
Files inventoried outside node_modules internals	557
First-party files (excluding tmp/node_modules)	179
First-party text files	160
First-party text lines	36,706
Model-relevant first-party text files	121
Direct dependencies	busboy 1.6.0

Executed verification

Command	Result	Interpretation
npm run check	PASS	Verified 29 pages, 102 source files, 7/12 Vercel functions, security headers, SEO metadata, local links and committed-secret rules.
npm audit --omit=dev --audit-level=high	PASS - 0 vulnerabilities	Applies to the two locked production dependencies at research time; not a model-worker SBOM.

Source families inspected

- Root configuration, README, package manifests, Vercel settings, GitHub workflows, OpenAPI YAML and environment documentation.
- All first-party JavaScript, HTML, CSS affecting detection/model UI, JSON/YAML, API handlers, worker code, gateway orchestration, persistence, validators and route layers.
- Supabase schema assumptions and persistence paths, model-performance UI, browser upload/crop behavior, direct APIs and v1 gateway APIs.
- All six PDFs, all 156 rendered pages, extracted text, diagrams, tables, model matrices, training/calibration/deployment/security sections and references.

The complete machine-readable inventory, hashes, line counts and relevance hits was retained as a temporary research artifact during generation. Product functionality was not modified.

6. PDFs and architecture sources analyzed

Source	Repository path	Pages	Review method
Architecture blueprint	inputs/pdfs/VeriTrust_Four_Core_Detection_Architecture_Blueprint_2026-08-09-1.pdf	22	Text extracted + every page rendered and visually inspected
Deepfake implementation handbook	output/pdfs/VeriTrust_Deepfake_Detection_Implementation_Handbook.pdf	26	Text extracted + every page rendered and visually inspected
Phishing implementation handbook	output/pdfs/VeriTrust_Phishing_Detection_Implementation_Handbook.pdf	25	Text extracted + every page rendered and visually inspected
Learning Module implementation handbook	output/pdfs/VeriTrust_Learning_Module_Implementation_Handbook.pdf	22	Text extracted + every page rendered and visually inspected
Link implementation handbook	output/pdfs/VeriTrust_Link_Detection_Implementation_Handbook.pdf	27	Text extracted + every page rendered and visually inspected
Unified Gateway implementation handbook	output/pdfs/VeriTrust_Unified_Gateway_Implementation_Handbook.pdf	34	Text extracted + every page rendered and visually inspected

Total: 156 pages. Page counts matched the supplied corpus. Visual review covered diagrams, architecture flows, callouts, matrices, source lists and current-versus-target statements; no PDF was assessed from text extraction alone.

Requirement ledger distilled before external search

Domain	Required scientific/evidence roles	Explicit non-model roles
Deepfake	Face manipulation; full-frame synthesis; independent residual/frequency forensic; trained OOD evidence	Face boxes/landmarks; decode; blur/compression/exposure/resolution quality
Phishing	Fast text; larger shadow; multilingual; structured-feature GBDT; optional bounded extraction	Header parsing; SPF/DKIM/DMARC; sender/relationship facts; URL extraction
Link	Four-class URL; binary/lexical shadow; structured URL GBDT; page text; screenshot embeddings	PSL/IDN/confusables; DNS/RDAP/TLS/HTTP; reputation; sandbox/browser collection
Gateway	No learned classifier initially	Expected-evidence ledger; deterministic versioned correlation; abstention/failure handling
Learning	No core HF inference requirement	Course/content/assessment/certification and analytics infrastructure

7. Current VeriTrust inference architecture

Today's executable path is configuration-driven and provider-first. It does not implement the handbook's immutable scientific registry or dedicated preprocessing workers.

Browser/UI	Direct API / v1 gateway routes	Detection service	Provider	Risk/evidence
image bytes / text / URL	validators + upload parsing	alias selection + fallback	HF raw bytes / HF JSON / Featherless chat	score normalization + thresholds + persistence

Transport contracts observed

Alias family	Current transport	Request	Response assumption	Timeout/payload
Pixel / Prism	HF hf-inference	Raw image bytes + content type	Nested/flat label-score arrays	20s default; direct 4 MiB, gateway 10 MiB
MailGuard	HF hf-inference	JSON {inputs, options.wait_for_model}	Text-classification score arrays	20s; direct text limits differ from gateway
Cortex	Featherless OpenAI-compatible	chat/completions messages	choices[0].message.content containing JSON	20s; 2 MiB provider response cap
Swift	HF hf-inference	JSON {inputs: normalized URL}	Four-class score arrays	20s; provider output cap 2 MiB
Sentinel	Unavailable/locked	None	None	Not runnable

Current-state findings that materially affect acquisition

- Deepfake normalization searches label text for real/fake and otherwise guesses from the top label or whether it contains '1'. Pixel and Prism silently fall back to one another even though target roles may differ.
- MailGuard hard-codes a four-class collapse: LABEL_0 and LABEL_2 are legitimate; LABEL_1 and LABEL_3 are phishing. A model with benign/phishing labels can normalize to phishing=1 because benign is not counted.
- Cortex is prompted to produce final label, confidence and risk, then guesses 'Phishing' if unstructured output contains that word. This exceeds the target bounded-extraction role.
- Swift maps LABEL_1 to phishing and LABEL_3 to defacement. The shortlisted URLBERT-v4 card states the opposite: LABEL_1=defacement and LABEL_3=phishing.
- Direct routes can fall back to local heuristics/provider substitutes; gateway model adapters disable those local fallbacks but still use the same normalization functions.
- Current risk weights and thresholds are provisional code constants, not qualified calibration artifacts. Raw scores are presented as probabilities/confidence.
- Gateway inference occurs before persistence resolves the active/canary model version, so recorded model-version metadata can diverge from the artifact that actually ran.

8. Current model aliases and environment configuration

Environment variable	Alias	Consumer/path	Current contract	Target scientific role
HF_DEEPPFAKE_PIXEL_MODEL	VeriTrust Pixel	lib/config.js -> deepfake_pixel -> detection service	Raw image binary HF classifier	Do not bind until alias is assigned one declared role; likely full-frame synthetic primary
HF_DEEPPFAKE_PRISM_MODEL	VeriTrust Prism	lib/config.js -> deepfake_prism -> detection service	Raw image binary HF classifier	Do not use as silent fallback; likely facial manipulation or independent synthetic shadow, not both
HF_PHISHING_MAILGUARD_MODEL	VeriTrust MailGuard	lib/config.js -> phishing_mailguard	HF JSON text classification	Low-latency text classifier with explicit label contract
HF_PHISHING_CORTEX_MODEL	VeriTrust Cortex	lib/config.js -> phishing_cortex	Featherless chat completion	Optional bounded observation extraction only; never final truth/risk
HF_LINK_SWIFT_MODEL	VeriTrust Swift	lib/config.js -> link_swift	HF JSON URL classification	Fast four-class URL model after label-map/worker refactor
No model variable	VeriTrust Sentinel	UI + link registry	Locked/unavailable	Rename/redefine as independent static feature model; do not imply a second HF model
HF_MODEL_PATHS	All aliases	JSON object merged by lib/config.js	Alias-keyed mutable owner/repo IDs	Replace with immutable registry references, processor and calibration versions
HF_MODEL_PATH	Legacy Pixel only	Fallback in lib/config.js	Single legacy owner/repo	Deprecate after registry migration
HF_TOKEN / HF_ACCESS_TOKEN	Provider auth	lib/config.js and provider clients	HF_TOKEN takes precedence	Use scoped secret; never treat token as artifact identity

Aliases are presentation/deployment names, not scientific model identities. If one alias cannot communicate a single stable task and input contract, change the alias or split it; do not force a scientifically different model underneath it.

9. Current-versus-target model architecture gap

Area	Current executable state	Target handbook state
Identity	Mutable environment ID resolved at request time	Immutable model + processor + code + calibration + benchmark record resolved before inference
Deepfake roles	Pixel/Prism interchangeable fallback	Separate face-manipulation, full-frame synthesis and independent residual evidence
Labels	String guessing and index assumptions	Registry-pinned explicit id2label; unknown output fails closed/abstains
Scores	Raw score treated as probability/confidence	Raw, normalized and calibrated values preserved separately
Preprocessing	Generic provider preprocessing	Versioned task-specific worker preprocessing and golden tensors
Failure	Provider/local heuristic substitution in direct routes	Typed unavailable/abstain; evidence-family completeness tracked
Gateway	Hard-coded default aliases and max/bonus correlation	Versioned deterministic policy over declared evidence; no learned meta-model initially
Persistence	Version resolved after inference	Registry selection and inference receipt bound before provider call
Calibration	Hard-coded 0.45/0.75/0.90-like thresholds	Versioned held-out calibration, ECE/Brier/reliability and cost-selected thresholds
OOD	Quality/rule heuristics can stand in for uncertainty	Separate trained embedding/OOD evidence and declared qualified distribution
URL parsing	Naive base-domain and class collapse	PSL-aware, IDN/confusable-safe, four-class and independent structured features
Security	Owner/repo pin only	Commit, file hashes, safe format, SBOM, mirror, license archive, conversion provenance

10. Model acquisition taxonomy

Class	Meaning	Count in investigated repositories	Decision rule
A	Download/benchmark directly	17	Task-specific artifact or embedding model suitable for controlled evaluation
B	Base model requiring VeriTrust fine-tuning	4	Useful backbone with no qualified task head
C	Custom VeriTrust model required	4 mandatory private families	No suitable public model or independence requires private training
D	Supporting non-HF component	Multiple	Deterministic preprocessing, YuNet, GBDT, PSL, sandbox, connectors
E	Reject	8	Wrong task/license/provenance/availability/supply-chain/deployment

Exactly one acquisition class is assigned to each of the 29 investigated Hub repositories. A shortlist flag is separate: it means controlled evaluation is justified. Critical license or task failures override scorecard averages.

11. Deepfake model requirements

Role	Scientific contract	Preferred acquisition	Boundary
A1 Face detection / landmarks	RGB/BGR image -> face boxes + 5-point landmarks + crop/alignment metadata	OpenCV Zoo YuNet or equivalent	Preprocessing only; never count as authenticity evidence
A2 Facial manipulation	Aligned/cropped face -> real/manipulated evidence	yemandy CLIP ViT-L/14 detector	Partial/local face changes; not full-frame synthesis
A3 Fully synthetic full frame	Decoded full image -> real/generated evidence	Community Forensics 384 primary; 224 latency alternative	Whole-image generator evidence; preserve source image
A4 Independent forensic	Residual/SRM/DCT/noise features -> forensic score	CUSTOM VERITRUST MODEL	Must create error diversity from semantic ViTs
A5 Quality	Decode/blur/compression/exposure/resolution/face sufficiency	Deterministic first; optional bounded ML for face quality	Quality gates, not OOD and not authenticity
A6 OOD	Pinned embeddings/features -> distance/density/conformal evidence	CUSTOM VERITRUST OOD HEAD	Qualified-distribution evidence with abstention

Image contract invariants

- Decode once to a canonical RGB image while preserving original bytes and metadata; reject invalid or oversized payloads before worker dispatch.
- Generate child artifacts explicitly: full frame, face crop, aligned face and residual views. Each evidence record must identify the artifact it analyzed.
- Never route a full-frame image into a face-manipulation model merely because both models use image classification. Never treat a face box as deepfake evidence.
- Pin resize, interpolation, crop/alignment, channel order, rescale and mean/std. A preprocessing change is a model-version change requiring requalification.
- Quality and OOD are separate. Blur/compression checks cannot establish whether the generator is outside the model's training distribution.

12. Deepfake Hugging Face research and shortlist

Repository	Role	Params	Repo size	Status	Class	Action
OwensLab/commfor-model-384	Full-frame synthetic-image primary	21.8M	83.22 MiB	VERIFIED - REQUIRES CUSTOM CODE	A	DOWNLOAD + SELF-HOST
OwensLab/commfor-model-224	Full-frame synthetic-image latency alternative	21.7M	82.66 MiB	VERIFIED - REQUIRES CUSTOM CODE	A	KEEP SHADOW ONLY
yermandy/deepfake-detection	Facial manipulation detector	UNKNOWN	1.70 GiB	VERIFIED - REQUIRES CUSTOM CODE	A	DOWNLOAD + SELF-HOST
openai/clip-vit-large-patch14	Facial-forensics backbone / embedding baseline	428M	12.76 GiB	VERIFIED - BASE MODEL ONLY	B	FINE-TUNE FIRST
buildborderless/CommunityForensics-Deepfake-Det-ViT	Synthetic-image standard-Transformers challenger	21.8M	1.26 GiB	VERIFIED - RESEARCH ONLY	A	BENCHMARK NOW
onnx-community/CommunityForensics-Deepfake-Det-ViT-ONNX	Community Forensics ONNX conversion	UNKNOWN	320.00 MiB	REJECTED	E	REJECT
dima806/deepfake_vs_real_image_detection	Legacy real-vs-fake image fallback	85.8M	3.84 GiB	VERIFIED - ACTIVE	A	KEEP SHADOW ONLY
capcheck/ai-image-detection	CIFAKE synthetic-image fallback	85.8M	327.32 MiB	VERIFIED - NO HF PROVIDER	A	KEEP SHADOW ONLY
prithivMLmods/Al-vs-Deepfake-vs-Real	Three-way image classifier	85.8M	1.92 GiB	REJECTED	E	REJECT

Preferred division of labor: yermandy for aligned facial manipulation; official Community Forensics 384 for full-frame synthesis; a private residual/frequency model for independence. The buildborderless integration is a useful reproducibility challenger, not an automatic substitute for the official checkpoint.

13. Deepfake detailed model dossiers

Exact live identity, artifacts, contract and decision; research record only.

Deepfake repository dossier 1/9

OwensLab/commfor-model-384

Decision	Acquisition	Readiness	Repository status	Current compatibility
DOWNLOAD + SELF-HOST	CLASS A	R1	VERIFIED - REQUIRES CUSTOM CODE	Dedicated worker only

Role	Full-frame synthetic-image primary
Repository	OwensLab/commfor-model-384
Pinned revision	6076002bf0d9dd37537f965ee2f06f826c333b61
Checked / modified	Checked 15 August 2026; Hub lastModified=2025-08-28T20:49:56.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	not declared / not declared
Architecture	patch_size=16
Parameters	21.8M (exact Safetensors metadata)
Repository storage	83.22 MiB
Primary artifacts	model.safetensors (83.22 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Official CVPR 2025 Community Forensics checkpoint. PyTorchModelHubMixin, not a generic HF image-classification contract. 384 input is the accuracy reference.	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files	Upstream: real=0, fake=1; exact Hub wrapper output must be fixture-verified

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	Modest GPU; CPU benchmark	FP32 83.21 MiB; FP16/BF16 41.60 MiB; INT8 20.80 MiB; INT4 10.40 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Community Forensics paper/card reports training across thousands of generators; report metrics are not reproduced here as qualified VeriTrust evidence.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	['OwensLab/CommunityForensics', 'OwensLab/CommunityForensics-Small']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Generator-held-out; manipulation-held-out; source/camera; screenshot/transcode; blur; compression; low resolution; face-quality slices; cross-dataset; OOD; FPR and selective-risk coverage.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: DOWNLOAD + SELF-HOST. This repository is VERIFIED - REQUIRES CUSTOM CODE and remains no higher than R1 from public evidence.

Deepfake repository dossier 2/9

OwensLab/commfor-model-224

Decision	Acquisition	Readiness	Repository status	Current compatibility
KEEP SHADOW ONLY	CLASS A	R1	VERIFIED - REQUIRES CUSTOM CODE	Dedicated worker only

Role	Full-frame synthetic-image latency alternative
Repository	OwensLab/commfor-model-224
Pinned revision	26afc31e6b40c312c3fd42c05a758be62446215b
Checked / modified	Checked 15 August 2026; Hub lastModified=2025-08-28T20:50:27.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	not declared / not declared
Architecture	patch_size=16
Parameters	21.7M (exact Safetensors metadata)
Repository storage	82.66 MiB
Primary artifacts	model.safetensors (82.66 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Official lower-resolution checkpoint from the same family and data as the 384 model. Useful for latency trade-off, not independent evidence.	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files	Upstream: real=0, fake=1; exact Hub wrapper output must be fixture-verified

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	CPU/modest GPU benchmark	FP32 82.65 MiB; FP16/BF16 41.32 MiB; INT8 20.66 MiB; INT4 10.33 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Same official study/family as 384. Treat any metric difference as a resolution trade-off to reproduce locally.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	['OwensLab/CommunityForensics', 'OwensLab/CommunityForensics-Small']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Generator-held-out; manipulation-held-out; source/camera; screenshot/transcode; blur; compression; low resolution; face-quality slices; cross-dataset; OOD; FPR and selective-risk coverage.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: KEEP SHADOW ONLY. This repository is VERIFIED - REQUIRES CUSTOM CODE and remains no higher than R1 from public evidence.

Deepfake repository dossier 3/9

yermandy/deepfake-detection

Decision	Acquisition	Readiness	Repository status	Current compatibility
DOWNLOAD + SELF-HOST	CLASS A	R1	VERIFIED - REQUIRES CUSTOM CODE	Preprocessed aligned-face worker only

Role	Facial manipulation detector
Repository	yermandy/deepfake-detection
Pinned revision	9a6857ec642deb57373c5437be803a199468b8c6
Checked / modified	Checked 15 August 2026; Hub lastModified=2025-04-01T10:35:32.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	image-classification / not declared
Architecture	Not fully declared in standard config; inspect pinned upstream runtime
Parameters	UNKNOWN (published/engineering estimate or UNKNOWN)
Repository storage	1.70 GiB
Primary artifacts	model.ckpt (1.13 GiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Official WACV 2026/2025 research implementation for partially manipulated faces using CLIP ViT-L/14. Requires DeepfakeBench-style preprocessing; the Hub repository is not a standard classifier.	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files	Binary real/manipulated logit semantics require upstream-code reproduction; HF config absent

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	Production GPU worker	UNKNOWN / not meaningful	unsafe weight files=1; repository Python files=0; Safetensors params=not reported

Evidence, licensing and qualification

Card reports cross-dataset video AUROC on Celeb-DF-v2, DFD, DFDC, FFIW and DeepSpeak v1; VeriTrust still needs image/face-level reproduction.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Generator-held-out; manipulation-held-out; source/camera; screenshot/transcode; blur; compression; low resolution; face-quality slices; cross-dataset; OOD; FPR and selective-risk coverage.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: DOWNLOAD + SELF-HOST. This repository is VERIFIED - REQUIRES CUSTOM CODE and remains no higher than R1 from public evidence.

Deepfake repository dossier 4/9

openai/clip-vit-large-patch14

Decision	Acquisition	Readiness	Repository status	Current compatibility
FINE-TUNE FIRST	CLASS B	R1	VERIFIED - BASE MODEL ONLY	Not a detector; processor/worker required

Role	Facial-forensics backbone / embedding baseline
Repository	openai/clip-vit-large-patch14
Pinned revision	32bd64288804d66eefd0ccbe215aa642df71cc41
Checked / modified	Checked 15 August 2026; Hub lastModified=2023-09-15T15:49:35.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	zero-shot-image-classification / transformers
Architecture	architectures=['CLIPModel']; model_type=clip
Parameters	428M (exact Safetensors metadata)
Repository storage	12.76 GiB
Primary artifacts	model.safetensors (1.59 GiB); pytorch_model.bin (1.59 GiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
General CLIP model, not a deepfake classifier. It cannot be assigned to Pixel/Prism without a VeriTrust task head and qualified preprocessing.	size=224; crop_size=224; do_resize=True; do_center_crop=True; do_normalize=True; image_mean=[0.48145466, 0.4578275, 0.40821073]; image_std=[0.26862954, 0.26130258, 0.27577711]; resample=3	No task label map - base/embedding model or UNKNOWN

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	16-24 GB GPU for tuning; GPU inference	FP32 1.59 GiB; FP16/BF16 815.61 MiB; INT8 407.81 MiB; INT4 203.90 MiB	unsafe weight files=1; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

No directly decision-grade published/card metric was found for this VeriTrust role; benchmark evidence remains insufficient.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
UNKNOWN	UNKNOWN / not declared	LICENSE UNKNOWN ON MODEL CARD - LEGAL REVIEW	Generator-held-out; manipulation-held-out; source/camera; screenshot/transcode; blur; compression; low resolution; face-quality slices; cross-dataset; OOD; FPR and selective-risk coverage.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: FINE-TUNE FIRST. This repository is VERIFIED - BASE MODEL ONLY and remains no higher than R1 from public evidence.

Deepfake repository dossier 5/9

buildborderless/CommunityForensics-DeepfakeDet-ViT

Decision	Acquisition	Readiness	Repository status	Current compatibility
BENCHMARK NOW	CLASS A	R2	VERIFIED - RESEARCH ONLY	Adapter refactor; single-logit sigmoid

Role	Synthetic-image standard-Transformers challenger
Repository	buildborderless/CommunityForensics-DeepfakeDet-ViT
Pinned revision	ac6ee457bea904a373065754107451793b56db00
Checked / modified	Checked 15 August 2026; Hub lastModified=2026-08-12T20:14:59.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	image-classification / transformers
Architecture	architectures=['ViTForImageClassification']; model_type=vit; num_hidden_layers=12; hidden_size=384; num_attention_heads=6; image_size=384; patch_size=16
Parameters	21.8M (exact Safetensors metadata)
Repository storage	1.26 GiB
Primary artifacts	model.safetensors (83.23 MiB); model_fixed.safetensors (83.23 MiB); model_legacy.safetensors (137.30 MiB); onnx/model.onnx (83.39 MiB); onnx/model_int8.onnx (21.96 MiB); +13 more model artifacts

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Third-party integration of Community Forensics corrected in July 2026. The card documents prior wrong weights/config and large quantization divergence on OOD inputs; equivalence to the official checkpoint must be verified.	size={'shortest_edge': 440}; crop_size=384; do_resize=True; do_center_crop=True; do_normalize=True; image_mean=[0.48145466, 0.4578275, 0.40821073]; image_std=[0.26862954, 0.26130258, 0.27577711]; resample=3; image_processor_type=CLIPImageProcessor	One logit; sigmoid(logit)=fake probability; real=1-fake

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
hf-inference: error / image-classification	CPU ONNX / modest GPU	FP32 83.21 MiB; FP16/BF16 41.60 MiB; INT8 20.80 MiB; INT4 10.40 MiB	unsafe weight files=2; repository Python files=6; Safetensors params=yes

Evidence, licensing and qualification

Card documents corrected July 2026 weights/config and warns INT8/Q4 can diverge by 10-70 percentage points on some OOD inputs.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Generator-held-out; manipulation-held-out; source/camera; screenshot/transcode; blur; compression; low resolution; face-quality slices; cross-dataset; OOD; FPR and selective-risk coverage.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: BENCHMARK NOW. This repository is VERIFIED - RESEARCH ONLY and remains no higher than R2 from public evidence.

Deepfake repository dossier 6/9

onnx-community/CommunityForensics-DeepfakeDet-ViT-ONNX

Decision	Acquisition	Readiness	Repository status	Current compatibility
REJECT	CLASS E	R0	REJECTED	Do not deploy

Role	Community Forensics ONNX conversion
Repository	onnx-community/CommunityForensics-DeepfakeDet-ViT-ONNX
Pinned revision	f9fc6516b37cd7d1bec3b94847b3682ce97249f0
Checked / modified	Checked 15 August 2026; Hub lastModified=2026-05-12T16:25:16.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	image-classification / transformers.js
Architecture	architectures=['ViTForImageClassification']; model_type=vit; num_hidden_layers=12; hidden_size=384; num_attention_heads=12; image_size=384; patch_size=16
Parameters	UNKNOWN (published/engineering estimate or UNKNOWN)
Repository storage	320.00 MiB
Primary artifacts	onnx/model.onnx (137.18 MiB); onnx/model_bnb4.onnx (23.29 MiB); onnx/model_fp16.onnx (68.77 MiB); onnx/model_int8.onnx (35.26 MiB); onnx/model_q4.onnx (23.29 MiB); +3 more model artifacts

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Conversion predates the July 2026 corrected integration and lacks an authoritative equivalence chain. Use a fresh export from a pinned verified checkpoint instead.	size={'height': 440, 'width': 440}; crop_size=384; do_resize=True; do_normalize=True; do_rescale=True; rescale_factor=0.00392156862745098; image_mean=[0.48145466, 0.4578275, 0.40821073]; image_std=[0.26862954, 0.26130258, 0.27577711]; resample=3; image_processor_type=ViTImageProcessor	No task label map - base/embedding model or UNKNOWN

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	None	UNKNOWN / not meaningful	unsafe weight files=0; repository Python files=0; Safetensors params=not reported

Evidence, licensing and qualification

No directly decision-grade published/card metric was found for this VeriTrust role; benchmark evidence remains insufficient.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	UNKNOWN / not declared	REJECT	Generator-held-out; manipulation-held-out; source/camera; screenshot/transcode; blur; compression; low resolution; face-quality slices; cross-dataset; OOD; FPR and selective-risk coverage.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: REJECT. This repository is REJECTED and remains no higher than R0 from public evidence.

Deepfake repository dossier 7/9

dima806/deepfake_vs_real_image_detection

Decision	Acquisition	Readiness	Repository status	Current compatibility
KEEP SHADOW ONLY	CLASS A	R2	VERIFIED - ACTIVE	Configuration-only for current raw-image adapter

Role	Legacy real-vs-fake image fallback
Repository	dima806/deepfake_vs_real_image_detection
Pinned revision	29e4cf9efc543845610045f6ba7e88e5cf9d9301
Checked / modified	Checked 15 August 2026; Hub lastModified=2025-01-04T09:44:17.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	image-classification / transformers
Architecture	architectures=["ViTForImageClassification"]; model_type=vit; num_hidden_layers=12; hidden_size=768; num_attention_heads=12; image_size=224; patch_size=16
Parameters	85.8M (exact Safetensors metadata)
Repository storage	3.84 GiB
Primary artifacts	checkpoint-28265/optimizer.pt (654.76 MiB); checkpoint-28265/pytorch_model.bin (327.37 MiB); checkpoint-28265/rng_state.pth (14.23 KiB); checkpoint-28265/scheduler.pt (627 B); checkpoint-28265/training_args.bin (3.93 KiB); +13 more model artifacts

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
The author explicitly warns of roughly three years of concept drift and recommends retraining. Live HF provider and clear Real/Fake labels make it useful only as a compatibility fixture, not the target primary.	size={'height': 224, 'width': 224}; do_resize=True; do_normalize=True; do_rescale=True; rescale_factor=0.00392156862745098; image_mean=[0.5, 0.5, 0.5]; image_std=[0.5, 0.5, 0.5]; resample=2; image_processor_type=ViTImageProcessor	0=Real, 1=Fake

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
hf-inference: live / image-classification	HF serverless currently live; CPU/GPU	FP32 327.30 MiB; FP16/BF16 163.65 MiB; INT8 81.83 MiB; INT4 40.91 MiB	unsafe weight files=15; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Card reports 0.9927 test accuracy but explicitly warns of severe concept drift and suggests retraining.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Generator-held-out; manipulation-held-out; source/camera; screenshot/transcode; blur; compression; low resolution; face-quality slices; cross-dataset; OOD; FPR and selective-risk coverage.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: KEEP SHADOW ONLY. This repository is VERIFIED - ACTIVE and remains no higher than R2 from public evidence.

Deepfake repository dossier 8/9

capcheck/ai-image-detection

Decision	Acquisition	Readiness	Repository status	Current compatibility
KEEP SHADOW ONLY	CLASS A	R2	VERIFIED - NO HF PROVIDER	Endpoint/worker refactor

Role	CIFAKE synthetic-image fallback
Repository	capcheck/ai-image-detection
Pinned revision	a6661e07d38f1a097bba07ca9415538819278f09
Checked / modified	Checked 15 August 2026; Hub lastModified=2025-12-06T03:52:02.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	image-classification / not declared
Architecture	architectures=['ViTForImageClassification']; model_type=vit; num_hidden_layers=12; hidden_size=768; num_attention_heads=12; image_size=224; patch_size=16
Parameters	85.8M (exact Safetensors metadata)
Repository storage	327.32 MiB
Primary artifacts	model.safetensors (327.32 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Safetensors-only derivative of a ViT-base CIFAKE model. Narrow generator/data coverage and high correlation with other ViT synthetic detectors limit its value.	size={'height': 224, 'width': 224}; do_resize=True; do_normalize=True; do_rescale=True; rescale_factor=0.00392156862745098; image_mean=[0.5, 0.5, 0.5]; image_std=[0.5, 0.5, 0.5]; resample=2; image_processor_type=ViTImageProcessor	0=REAL, 1=FAKE

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	Self-hosted Transformers / ONNX export	FP32 327.30 MiB; FP16/BF16 163.65 MiB; INT8 81.83 MiB; INT4 40.91 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Card reports CIFAKE-based binary task; no generator-held-out VeriTrust evidence.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	['CIFAKE']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Generator-held-out; manipulation-held-out; source/camera; screenshot/transcode; blur; compression; low resolution; face-quality slices; cross-dataset; OOD; FPR and selective-risk coverage.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: KEEP SHADOW ONLY. This repository is VERIFIED - NO HF PROVIDER and remains no higher than R2 from public evidence.

Deepfake repository dossier 9/9

prithivMLmods/Al-vs-Deepfake-vs-Real

Decision	Acquisition	Readiness	Repository status	Current compatibility
REJECT	CLASS E	R0	REJECTED	Current binary collapse discards Artificial class

Role	Three-way image classifier
Repository	prithivMLmods/Al-vs-Deepfake-vs-Real
Pinned revision	e1df5230c067ff4b189230343e548c365d80bbdb
Checked / modified	Checked 15 August 2026; Hub lastModified=2025-02-23T08:54:13.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	image-classification / transformers
Architecture	architectures=['ViTForImageClassification']; model_type=vit; num_hidden_layers=12; hidden_size=768; num_attention_heads=12; image_size=224; patch_size=16
Parameters	85.8M (exact Safetensors metadata)
Repository storage	1.92 GiB
Primary artifacts	checkpoint-564/model.safetensors (327.33 MiB); checkpoint-564/optimizer.pt (654.77 MiB); checkpoint-564/rng_state.pth (13.91 KiB); checkpoint-564/scheduler.pt (1.04 KiB); checkpoint-564/training_args.bin (5.18 KiB); +7 more model artifacts

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
The card itself says it is not a deepfake detector; its Artificial/Deepfake/Real taxonomy merges distinct VeriTrust roles and current code would normalize it incorrectly.	size={'height': 224, 'width': 224}; do_resize=True; do_normalize=True; do_rescale=True; rescale_factor=0.00392156862745098; image_mean=[0.5, 0.5, 0.5]; image_std=[0.5, 0.5, 0.5]; resample=2; image_processor_type=ViTImageProcessor	0=Artificial, 1=Deepfake, 2=Real

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	None	FP32 327.30 MiB; FP16/BF16 163.65 MiB; INT8 81.83 MiB; INT4 40.91 MiB	unsafe weight files=9; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Card reports 0.9750 accuracy on a 4,000-item evaluation; provenance and split design are insufficient for target roles.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	['prithivMLmods/Al-vs-Deepfake-vs-Real']	REJECT PENDING DATA PROVENANCE	Generator-held-out; manipulation-held-out; source/camera; screenshot/transcode; blur; compression; low resolution; face-quality slices; cross-dataset; OOD; FPR and selective-risk coverage.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: REJECT. This repository is REJECTED and remains no higher than R0 from public evidence.

14. Phishing model requirements

Role	Contract	Acquisition	Boundary
B1 Fast text classifier	Subject + new body, optionally bounded context; binary phishing evidence	50-100M English model on CPU	No header/auth truth unless provided as structured features
B2 Larger shadow	Same semantic text artifact; independent architecture/data where possible	ModernBERT-large shadow	Shadow only until leakage-safe benefit is shown
B3 Multilingual	Locale-aware message text; English/Hinglish and declared locales	Fine-tune mDeBERTa-v3-base	Public base model has no phishing head
B4 Structured features	SPF/DKIM/DMARC, sender/domain/relationship/content features	Private LightGBM/XGBoost/ONNX	Independent of Transformer token representation
B5 Bounded extraction	Credential/payment/urgency/impersonation observations + evidence spans	Optional Mistral 7B lab only	Never final truth, risk tier, action or enforcement

Text construction

- Use an explicit framing such as [SUBJECT]...[/SUBJECT][BODY]...[/BODY] only if the training recipe uses the same tokens; otherwise preserve the model's documented format.
- Separate newly written body from quoted history and signatures where possible; keep flags indicating removed/retained material.
- Pin tokenizer files, Unicode normalization, URL/email masking policy, maximum tokens, truncation side, padding and chunk aggregation.
- For long messages, benchmark head+tail and sliding chunks. Never silently take the first 512 tokens as a universal production rule.
- Do not concatenate privileged headers or authentication facts into a public provider prompt; structured features belong in a separate controlled model.

15. Phishing Hugging Face research and shortlist

Repository	Role	Params	Provider	Status	Classes	Action
cybersectonv/phishing-email-detection-distilbert-v2.4.1	Mixed email/URL four-class compatibility candidate	67M	hf-inference: live / text-classification	VERIFIED - ACTIVE	A	BENCHMARK NOW
dima806/phishing-email-detection	Binary DistilBERT fallback	65.8M	No live Inference Provider mapping	VERIFIED - NO HF PROVIDER	A	KEEP SHADOW ONLY
aamoshdahal/email-phishing-distilbert-finetuned	Binary DistilBERT candidate	67M	No live Inference Provider mapping	REJECTED	E	REJECT
ealvaradob/bert-finetuned-phishing	Large English shadow classifier	336M	hf-inference: live / text-classification	VERIFIED - ACTIVE	A	KEEP SHADOW ONLY
microsoft/mdeberta-v3-base	Preferred multilingual fine-tuning backbone	279M	hf-inference: live / fill-mask	VERIFIED - BASE MODEL ONLY	B	FINE-TUNE FIRST
mistralai/Mistral-7B-Instruct-v0.3	Optional bounded observation extractor	7.25B	novita: error / conversational	VERIFIED - NO HF PROVIDER	A	KEEP DISABLED / SELF-HOST LAB
FacebookAI/xlm-roberta-base	Multilingual fine-tuning fallback	279M	hf-inference: live / fill-mask	VERIFIED - BASE MODEL ONLY	B	FINE-TUNE FIRST
Sonje03/phishlens-distilbert	Preferred fast English classifier candidate	67M	No live Inference Provider mapping	VERIFIED - NO HF PROVIDER	A	DOWNLOAD + SELF-HOST
mikaelnurminen/modernbert-large-phishing	Large shadow/error-diversity classifier	396M	No live Inference Provider mapping	VERIFIED - NO HF PROVIDER	A	BENCHMARK SHADOW
JustAPR/email-phishing-distilbert-finetuned	Copied binary DistilBERT card	67M	No live Inference Provider mapping	REJECTED	E	REJECT
Qwen/Qwen2.5-3B-Instruct	Optional bounded extractor alternative	3.09B	featherless-ai: live / conversational	REJECTED	E	REJECT

Preferred practical stack: Sonje03 as the fast English acquisition candidate; mDeBERTa-v3-base for VeriTrust multilingual fine-tuning; ModernBERT-large as a shadow with its eager-attention constraint; a private structured-feature GBDT. The optional LLM remains disabled unless bounded extraction adds measurable value.

16. Phishing detailed model dossiers

Task head, label map, modality, provider, language and leakage evidence; research record only.

Phishing repository dossier 1/11

cybersectony/phishing-email-detection-distilbert_v2.4.1

Decision	Acquisition	Readiness	Repository status	Current compatibility
BENCHMARK NOW	CLASS A	R2	VERIFIED - ACTIVE	Current MailGuard label collapse matches card mapping

Role	Mixed email/URL four-class compatibility candidate
Repository	cybersectony/phishing-email-detection-distilbert_v2.4.1
Pinned revision	6724dec17360ab1f6450553f1d950404f66a5dde
Checked / modified	Checked 15 August 2026; Hub lastModified=2024-10-28T10:02:55.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / transformers
Architecture	architectures=["DistilBertForSequenceClassification"]; model_type=distilbert; max_position_embeddings=512
Parameters	67M (exact Safetensors metadata)
Repository storage	255.43 MiB
Primary artifacts	model.safetensors (255.43 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
The card calls this multilabel but examples use a four-class softmax. Classes mix email and URL modalities; it is not a clean binary email classifier and must be campaign/time benchmarked.	tokenizer_class=DistilBertTokenizer; model_max_length=512	0=LABEL_0, 1=LABEL_1, 2=LABEL_2, 3=LABEL_3

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
hf-inference: live / text-classification	HF serverless currently live; CPU	FP32 255.42 MiB; FP16/BF16 127.71 MiB; INT8 63.85 MiB; INT4 31.93 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Card reports ~99.58% aggregate metrics on a mixed email/URL dataset; campaign/time/identity leakage controls are undocumented.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	["cybersectony/PhishingEmailDetectionv2.0"]	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: BENCHMARK NOW. This repository is VERIFIED - ACTIVE and remains no higher than R2 from public evidence.

Phishing repository dossier 2/11

dima806/phishing-email-detection

Decision	Acquisition	Readiness	Repository status	Current compatibility
KEEP SHADOW ONLY	CLASS A	R1	VERIFIED - NO HF PROVIDER	Label-map and endpoint refactor

Role	Binary DistilBERT fallback
Repository	dima806/phishing-email-detection
Pinned revision	310a5602280486e01dd60e37230980cfcc327b41
Checked / modified	Checked 15 August 2026; Hub lastModified=2024-10-19T11:03:28.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / transformers
Architecture	architectures=['DistilBertForSequenceClassification']; model_type=distilbert; max_position_embeddings=512
Parameters	65.8M (exact Safetensors metadata)
Repository storage	2.45 GiB
Primary artifacts	model.safetensors (250.95 MiB); pytorch_model.bin (250.98 MiB); training_args.bin (3.50 KiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Clear config labels but an extremely sparse card delegates evidence to Kaggle. No live HF provider and limited provenance make it a low-priority fixture.	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files	0=SAVE EMAIL, 1=PHISHING EMAIL

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	Self-hosted CPU	FP32 250.94 MiB; FP16/BF16 125.47 MiB; INT8 62.74 MiB; INT4 31.37 MiB	unsafe weight files=2; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

No metrics in the Hub card; external Kaggle notebook is referenced. Hub evidence is insufficient.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: KEEP SHADOW ONLY. This repository is VERIFIED - NO HF PROVIDER and remains no higher than R1 from public evidence.

Phishing repository dossier 3/11

aamoshdahal/email-phishing-distilbert-finetuned

Decision	Acquisition	Readiness	Repository status	Current compatibility
REJECT	CLASS E	R0	REJECTED	Missing config label map and no provider

Role	Binary DistilBERT candidate
Repository	aamoshdahal/email-phishing-distilbert-finetuned
Pinned revision	f9d211125fcb6e85cf73630957667bc2a6e93ed3
Checked / modified	Checked 15 August 2026; Hub lastModified=2025-04-02T00:45:47.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / transformers
Architecture	architectures=['DistilBertForSequenceClassification']; model_type=distilbert; max_position_embeddings=512
Parameters	67M (exact Safetensors metadata)
Repository storage	255.43 MiB
Primary artifacts	model.safetensors (255.43 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Task fit is plausible, but the Hub repository declares no license and config labels are generic. Do not acquire for product use until the owner supplies explicit licensing and a pinned label contract.	tokenizer_class=DistilBertTokenizer; model_max_length=512	Card says 0=legitimate, 1=phishing; config is generic - NOT IMPLEMENTATION READY

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	None	FP32 255.41 MiB; FP16/BF16 127.71 MiB; INT8 63.85 MiB; INT4 31.93 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Card reports internal epoch metrics and a separate 96.62% test summary; split/campaign leakage must be independently assessed.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
UNKNOWN	UNKNOWN / not declared	LICENSE UNKNOWN - REJECT	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: REJECT. This repository is REJECTED and remains no higher than R0 from public evidence.

Phishing repository dossier 4/11

ealvaradob/bert-finetuned-phishing

Decision	Acquisition	Readiness	Repository status	Current compatibility
KEEP SHADOW ONLY	CLASS A	R2	VERIFIED - ACTIVE	MailGuard benign-label fix required

Role	Large English shadow classifier
Repository	ealvaradob/bert-finetuned-phishing
Pinned revision	fa8fb73a007174c410ab7160d4e4c6e6b8d998d4
Checked / modified	Checked 15 August 2026; Hub lastModified=2024-02-07T05:11:47.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / transformers
Architecture	architectures=['BertForSequenceClassification']; model_type=bert; num_hidden_layers=24; hidden_size=1024; num_attention_heads=16; max_position_embeddings=512
Parameters	336M (published/engineering estimate or UNKNOWN)
Repository storage	39.96 GiB
Primary artifacts	pytorch_model.bin (1.25 GiB); training_args.bin (4.43 KiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
BERT-large task-specific classifier across URL/email/SMS/web text. Current normalizer recognizes phishing but not benign, causing pathological normalization when full scores are returned. Repository storage includes many redundant checkpoints.	tokenizer_class=BertTokenizer; model_max_length=512	0=benign, 1=phishing

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
hf-inference: live / text-classification	HF serverless / GPU or strong CPU	FP32 1.25 GiB; FP16/BF16 640.87 MiB; INT8 320.43 MiB; INT4 160.22 MiB	unsafe weight files=2; repository Python files=0; Safetensors params=not reported

Evidence, licensing and qualification

Card reports 0.9717 accuracy and 0.0249 FPR over mixed URL/email/SMS/web content.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	['ealvaradob/phishing-dataset']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: KEEP SHADOW ONLY. This repository is VERIFIED - ACTIVE and remains no higher than R2 from public evidence.

Phishing repository dossier 5/11

microsoft/mdeberta-v3-base

Decision	Acquisition	Readiness	Repository status	Current compatibility
FINE-TUNE FIRST	CLASS B	R1	VERIFIED - BASE MODEL ONLY	No classification head; new worker

Role	Preferred multilingual fine-tuning backbone
Repository	microsoft/mdeberta-v3-base
Pinned revision	a0484667b22365f84929a935b5e50a51f71f159d
Checked / modified	Checked 15 August 2026; Hub lastModified=2023-04-06T05:32:33.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	fill-mask / transformers
Architecture	model_type=deberta-v2; num_hidden_layers=12; hidden_size=768; num_attention_heads=12; max_position_embeddings=512
Parameters	279M (published/engineering estimate or UNKNOWN)
Repository storage	5.45 GiB
Primary artifacts	pytorch_model.bin (1.24 GiB); pytorch_model.generator.bin (905.32 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Multilingual masked-language-model backbone, not a phishing classifier. Recommended practical backbone for a VeriTrust locale-aware head after dataset and tokenizer requalification.	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files	No task label map - base/embedding model or UNKNOWN

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
hf-inference: live / fill-mask	Modest/production GPU for tuning; CPU/GPU inference	FP32 1.04 GiB; FP16/BF16 532.15 MiB; INT8 266.08 MiB; INT4 133.04 MiB	unsafe weight files=2; repository Python files=0; Safetensors params=not reported

Evidence, licensing and qualification

No directly decision-grade published/card metric was found for this VeriTrust role; benchmark evidence remains insufficient.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: FINE-TUNE FIRST. This repository is VERIFIED - BASE MODEL ONLY and remains no higher than R1 from public evidence.

mistralai/Mistral-7B-Instruct-v0.3

Role	Optional bounded observation extractor
Repository	mistralai/Mistral-7B-Instruct-v0.3
Pinned revision	c170c708c41dac9275d15a8fff4eca08d52bab71
Checked / modified	Checked 15 August 2026; Hub lastModified=2025-12-03T12:13:48.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	not declared / vllm
Architecture	architectures=['MistralForCausalLM']; model_type=mistral; num_hidden_layers=32; hidden_size=4096; num_attention_heads=32; max_position_embeddings=32768
Parameters	7.25B (exact Safetensors metadata)
Repository storage	27.00 GiB
Primary artifacts	consolidated.safetensors (13.50 GiB); model-00001-of-00003.safetensors (4.61 GiB); model-00002-of-00003.safetensors (4.66 GiB); model-00003-of-00003.safetensors (4.23 GiB)

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Apache-2.0 general instruct model, not a phishing truth model. The current live provider mapping shows Novita status error. Use only for bounded structured observations if privacy, JSON conformance and injection tests justify it.	tokenizer_class=LlamaTokenizer; model_max_length=1000000000000000019884624838656	No task label map - base/embedding model or UNKNOWN

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
novita: error / conversational	16-24 GB GPU; self-host	FP32 27.00 GiB; FP16/BF16 13.50 GiB; INT8 6.75 GiB; INT4 3.38 GiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Disposition: KEEP DISABLED / SELF-HOST LAB. This repository is VERIFIED - NO HF PROVIDER and remains no higher than R1 from public evidence.

Phishing repository dossier 7/11

FacebookAI/xlm-roberta-base

Decision	Acquisition	Readiness	Repository status	Current compatibility
FINE-TUNE FIRST	CLASS B	R1	VERIFIED - BASE MODEL ONLY	No classification head; new worker

Role	Multilingual fine-tuning fallback
Repository	FacebookAI/xlm-roberta-base
Pinned revision	e73636d4f797dec63c3081bb6ed5c7b0bb3f2089
Checked / modified	Checked 15 August 2026; Hub lastModified=2024-02-19T12:48:21.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	fill-mask / transformers
Architecture	architectures=['XLMRobertaForMaskedLM']; model_type=xlm-roberta; num_hidden_layers=12; hidden_size=768; num_attention_heads=12; max_position_embeddings=514
Parameters	279M (exact Safetensors metadata)
Repository storage	5.92 GiB
Primary artifacts	model.onnx (1.75 GiB); model.safetensors (1.04 GiB); pytorch_model.bin (1.04 GiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Mature 100-language backbone. It is a fallback if mDeBERTa-v3 tokenization or locale benchmarks are inferior; not directly assignable to MailGuard.	model_max_length=512	No task label map - base/embedding model or UNKNOWN

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
hf-inference: live / fill-mask	Modest GPU tuning; CPU/GPU inference	FP32 1.04 GiB; FP16/BF16 531.93 MiB; INT8 265.97 MiB; INT4 132.98 MiB	unsafe weight files=1; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

No directly decision-grade published/card metric was found for this VeriTrust role; benchmark evidence remains insufficient.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: FINE-TUNE FIRST. This repository is VERIFIED - BASE MODEL ONLY and remains no higher than R1 from public evidence.

Phishing repository dossier 8/11

Sonje03/phishlens-distilbert

Decision	Acquisition	Readiness	Repository status	Current compatibility
DOWNLOAD + SELF-HOST	CLASS A	R2	VERIFIED - NO HF PROVIDER	Label-map and self-host worker required

Role	Preferred fast English classifier candidate
Repository	Sonje03/phishlens-distilbert
Pinned revision	b585a2b9bbf6777b344d017f4eaff9423363a489
Checked / modified	Checked 15 August 2026; Hub lastModified=2026-06-14T22:46:46.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / transformers
Architecture	architectures=['DistilBertForSequenceClassification']; model_type=distilbert; max_position_embeddings=512
Parameters	67M (exact Safetensors metadata)
Repository storage	255.43 MiB
Primary artifacts	model.safetensors (255.43 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Honest English-only binary card, explicit Safe=0/Phishing=1 and 256-token recipe, with multi-source corpus and limitations. Preferred fast candidate for controlled evaluation, not production on card metrics.	tokenizer_class=BertTokenizer; model_max_length=512	0=Safe, 1=Phishing

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	Standard CPU worker	FP32 255.41 MiB; FP16/BF16 127.71 MiB; INT8 63.85 MiB; INT4 31.93 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Card reports 97.34% accuracy on 29,555 emails and explicitly says English-only and not sufficient alone for production defense.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: DOWNLOAD + SELF-HOST. This repository is VERIFIED - NO HF PROVIDER and remains no higher than R2 from public evidence.

Phishing repository dossier 9/11

mikaelnurminen/modernbert-large-phishing

Decision	Acquisition	Readiness	Repository status	Current compatibility
BENCHMARK SHADOW	CLASS A	R2	VERIFIED - NO HF PROVIDER	Dedicated worker; never current adapter

Role	Large shadow/error-diversity classifier
Repository	mikaelnurminen/modernbert-large-phishing
Pinned revision	d28c16f882454115d71375d3321fbfd90e566490
Checked / modified	Checked 15 August 2026; Hub lastModified=2026-02-27T11:53:55.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / transformers
Architecture	architectures=['ModernBertForSequenceClassification']; model_type=modernbert; num_hidden_layers=28; hidden_size=1024; num_attention_heads=16; max_position_embeddings=8192
Parameters	396M (exact Safetensors metadata)
Repository storage	4.20 GiB
Primary artifacts	model.safetensors (755.01 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Strong card transparency: real-world holdout degradation, threshold shift and NaN risk with SDPA batching are documented. Use eager attention and batch=1 until reproduced.	tokenizer_class=PreTrainedTokenizer; model_max_length=8192	0=benign, 1=phishing

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	Modest GPU; eager attention	FP32 1.47 GiB; FP16/BF16 754.99 MiB; INT8 377.50 MiB; INT4 188.75 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Card reports internal F1 0.8822 and real-world holdout F1 0.7292 at threshold 0.47; best holdout threshold shifts to 0.03.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: BENCHMARK SHADOW. This repository is VERIFIED - NO HF PROVIDER and remains no higher than R2 from public evidence.

Phishing repository dossier 10/11

JustAPR/email-phishing-distilbert-finetuned

Decision	Acquisition	Readiness	Repository status	Current compatibility
REJECT	CLASS E	R0	REJECTED	Card code references another repository

Role	Copied binary DistilBERT card
Repository	JustAPR/email-phishing-distilbert-finetuned
Pinned revision	9d8e0ac6797de33a3c97e2fc13fe8f99b1b12dfc
Checked / modified	Checked 15 August 2026; Hub lastModified=2026-02-25T18:21:19.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / transformers
Architecture	architectures=['DistilBertForSequenceClassification']; model_type=distilbert; max_position_embeddings=512
Parameters	67M (exact Safetensors metadata)
Repository storage	255.43 MiB
Primary artifacts	model.safetensors (255.43 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Repository owner differs from the model ID throughout the copied card; config lacks a verified label map and no license is declared. Supply-chain/provenance failure.	tokenizer_class=DistilBertTokenizer; model_max_length=512	Card says 0=legitimate, 1=phishing; config/card ownership conflict

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	None	FP32 255.41 MiB; FP16/BF16 127.71 MiB; INT8 63.85 MiB; INT4 31.93 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Copied card reports strong metrics but references another owner/model and lacks a trustworthy local identity chain.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
UNKNOWN	UNKNOWN / not declared	LICENSE UNKNOWN - REJECT	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: REJECT. This repository is REJECTED and remains no higher than R0 from public evidence.

Phishing repository dossier 11/11

Qwen/Qwen2.5-3B-Instruct

Decision	Acquisition	Readiness	Repository status	Current compatibility
REJECT	CLASS E	R0	REJECTED	Featherless live, but role/prompt refactor still required

Role	Optional bounded extractor alternative
Repository	Qwen/Qwen2.5-3B-Instruct
Pinned revision	aa8e72537993ba99e69dfaafa59ed015b17504d1
Checked / modified	Checked 15 August 2026; Hub lastModified=2024-09-25T12:33:00.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-generation / transformers
Architecture	architectures=["Qwen2ForCausalLM"]; model_type=qwen2; num_hidden_layers=36; hidden_size=2048; num_attention_heads=16; max_position_embeddings=32768
Parameters	3.09B (exact Safetensors metadata)
Repository storage	5.75 GiB
Primary artifacts	model-00001-of-00002.safetensors (3.70 GiB); model-00002-of-00002.safetensors (2.05 GiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Technically attractive 3.09B JSON-capable model with a live Featherless provider, but its Qwen Research License permits non-commercial use only unless a separate commercial license is obtained.	tokenizer_class=Qwen2Tokenizer; model_max_length=131072	No task label map - base/embedding model or UNKNOWN

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
featherless-ai: live / conversational	None for commercial product	FP32 11.50 GiB; FP16/BF16 5.75 GiB; INT8 2.87 GiB; INT4 1.44 GiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

No directly decision-grade published/card metric was found for this VeriTrust role; benchmark evidence remains insufficient.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
other	UNKNOWN / not declared	NON-COMMERCIAL - REJECT	Campaign/time/sender/domain/tenant-separated; English, Hinglish and relevant locales; BEC/financial fraud/smishing; quoted-thread contamination; legitimate urgency; FPR and calibration.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: REJECT. This repository is REJECTED and remains no higher than R0 from public evidence.

17. Link and malicious-URL model requirements

Role	Contract	Acquisition	Boundary
C1 Fast multiclass URL	Raw/canonical URL -> benign/phishing/malware/defacement evidence	URLBERT-tiny-v4 four-class	Explicit card map; preserve multiclass probabilities
C2 Binary URL shadow	URL -> phishing/benign evidence	pirocheto ONNX preferred for diversity; URLBERT binary as correlated comparison	Shadow correlation measured, not assumed
C3 Static feature model	PSL/IDN/confusable/lexical/encoding/relationship features	Private GBDT/ONNX	Independent non-Transformer evidence
C4 Sanitized page text	Sandbox title/visible text/form labels only	Fine-tune DeBERTa-v3-small	Async; no scripts, secrets or direct network in model worker
C5 Screenshot/brand	Screenshot -> embedding similarity to approved references	SigLIP2 base	Evidence only; similarity is not maliciousness
C6 Connectors	Reputation/DNS/RDAP/TLS/HTTP/browser facts	Non-HF services and collectors	Never describe as models

URL normalization contract

- Preserve the original URL and a separately derived display/canonical form. Do not decode or recase blindly; percent encoding and Unicode can be evidence.
- Use Public Suffix List interpretation for registrable domains; handle special-use IP/address forms, IPv6, userinfo, IDNA and mixed-script confusables.
- Record exactly which string the model saw. A tokenizer trained on raw URLs may be invalidated by normalization that improves a different model.
- Pin maximum tokens/characters and truncation behavior. URLBERT-v4 has a 64-position configuration; long-tail truncation must be benchmarked.
- Malformed percent escapes must produce typed validation output, not an uncaught decode error or silent normalization change.

18. Link Hugging Face research and shortlist

Repository	Role	Params	Provider	Status	Class	Action
CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	Preferred fast four-class URL candidate	3.69M	No live Inference Provider mapping	VERIFIED - NO HF PROVIDER	A	DOWNLOAD + SELF-HOST
CrabInHoney/urlbert-tiny-v4-phishing-classifier	Binary phishing URL shadow	3.69M	No live Inference Provider mapping	VERIFIED - NO HF PROVIDER	A	KEEP SHADOW ONLY
CrabInHoney/urlbert-tiny-v5	New URL encoder + task-head challenger	58.3M	No live Inference Provider mapping	VERIFIED - REQUIRES CUSTOM CODE	A	BENCHMARK AFTER V4
elftsdmr/malware-url-detect	Turkey-focused URL classifier	178M	No live Inference Provider mapping	REJECTED	E	REJECT
microsoft/deberta-v3-small	Sanitized page-text fine-tuning backbone	142M	hf-inference: live / fill-mask	VERIFIED - BASE MODEL ONLY	B	FINE-TUNE FIRST
google/siglip2-base-patch16-224	Screenshot/approved-brand embedding model	375M	No live Inference Provider mapping	VERIFIED - NO HF PROVIDER	A	DOWNLOAD + SELF-HOST
protectai/codebert-base-Malicious_URLs-onnx	CodeBERT ONNX malicious URL model	125M	No live Inference Provider mapping	VERIFIED - ARCHIVED/STALE	E	REJECT
pirocheto/phishing-url-detection	Independent linear-SVM phishing shadow	UNKNOWN	No live Inference Provider mapping	VERIFIED - NO HF PROVIDER	A	DOWNLOAD ONNX + BENCHMARK
DiligentAI/urlbert-url-classifier	Official-site vs platform URL classifier	111M	No live Inference Provider mapping	REJECTED	E	REJECT

URLBERT-v4 remains the simplest acquisition-ready four-class checkpoint, but it requires self-hosting and a corrected label map. URLBERT-v5 is a newer encoder/head research challenger, not a standard classifier. The independent production secondary should be a private structured-feature GBDT; pirocheto ONNX is a useful external lexical shadow.

19. Link detailed model dossiers

Raw URL, page-text and screenshot evidence remain distinct from connectors; research record only.

Link repository dossier 1/9

CrabInHoney/urlbert-tiny-v4-malicious-url-classifier

Decision	Acquisition	Readiness	Repository status	Current compatibility
DOWNLOAD + SELF-HOST	CLASS A	R2	VERIFIED - NO HF PROVIDER	Label-map swap and self-host worker required

Role	Preferred fast four-class URL candidate
Repository	CrabInHoney/urlbert-tiny-v4-malicious-url-classifier
Pinned revision	917ded0543a630d6e82570bb01ae692f6cbb95f1
Checked / modified	Checked 15 August 2026; Hub lastModified=2026-01-26T11:43:47.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / transformers
Architecture	architectures=["BertForSequenceClassification"]; model_type=bert; num_hidden_layers=8; hidden_size=192; num_attention_heads=8; max_position_embeddings=64
Parameters	3.69M (exact Safetensors metadata)
Repository storage	14.08 MiB
Primary artifacts	model.safetensors (14.08 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Exact map is LABEL_0 benign, LABEL_1 defacement, LABEL_2 malware, LABEL_3 phishing. Current VeriTrust swaps LABEL_1 and LABEL_3, so dropping this ID into Swift would misstate the threat class.	tokenizer_class=BertTokenizer; model_max_length=64	0=LABEL_0, 1=LABEL_1, 2=LABEL_2, 3=LABEL_3

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	Standard CPU worker / ONNX export	FP32 14.06 MiB; FP16/BF16 7.03 MiB; INT8 3.52 MiB; INT4 1.76 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Card reports 0.9922 accuracy / 0.9734 phishing F1 on the Malicious URLs Dataset; temporal/domain grouping is not documented.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Domain/campaign/time-grouped; newly registered and compromised legitimate domains; shorteners; IDN/confusables; malformed/encoded URLs; class-specific phishing/malware/defacement; India lures; false positives per 10k.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: DOWNLOAD + SELF-HOST. This repository is VERIFIED - NO HF PROVIDER and remains no higher than R2 from public evidence.

[Link repository dossier 2/9](#)

CrabInHoney/urlbert-tiny-v4-phishing-classifier

Decision	Acquisition	Readiness	Repository status	Current compatibility
KEEP SHADOW ONLY	CLASS A	R2	VERIFIED - NO HF PROVIDER	Self-host worker; explicit label map

Role	Binary phishing URL shadow
Repository	CrabInHoney/urlbert-tiny-v4-phishing-classifier
Pinned revision	fd962a5cd04e20ceed46aa8a6e31a2b40d4c8916
Checked / modified	Checked 15 August 2026; Hub lastModified=2026-01-26T11:42:35.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / not declared
Architecture	architectures=['BertForSequenceClassification']; model_type=bert; num_hidden_layers=8; hidden_size=192; num_attention_heads=8; max_position_embeddings=64
Parameters	3.69M (exact Safetensors metadata)
Repository storage	14.08 MiB
Primary artifacts	model.safetensors (14.08 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Very small binary URL classifier. Same owner, tiny BERT family and overlapping data as the multiclass primary imply high error correlation; it cannot satisfy independent-secondary requirements by itself.	tokenizer_class=BertTokenizer; model_max_length=64	LABEL_0=good/safe; LABEL_1=phishing ('fish' in card example)

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	Standard CPU worker / ONNX export	FP32 14.06 MiB; FP16/BF16 7.03 MiB; INT8 3.52 MiB; INT4 1.76 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

Card reports v4 comparison metrics on ealvaradob/phishing-dataset; same-family correlation remains high.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	['ealvaradob/phishing-dataset']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Domain/campaign/time-grouped; newly registered and compromised legitimate domains; shorteners; IDN/confusables; malformed/encoded URLs; class-specific phishing/malware/defacement; India lures; false positives per 10k.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: KEEP SHADOW ONLY. This repository is VERIFIED - NO HF PROVIDER and remains no higher than R2 from public evidence.

CrabInHoney/urlbert-tiny-v5

Role	New URL encoder + task-head challenger
Repository	CrabInHoney/urlbert-tiny-v5
Pinned revision	043ab17548c371ef8a0ce6d3f7c728e48f844e7d
Checked / modified	Checked 15 August 2026; Hub lastModified=2026-05-08T14:49:22.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	feature-extraction / transformers
Architecture	architectures=["DistilBertModel"]; model_type=distilbert; max_position_embeddings=512
Parameters	58.3M (exact Safetensors metadata)
Repository storage	292.77 MiB
Primary artifacts	heads/MSMalicious-URLs-dataset_head.safetensors (2.28 MiB); heads/PSSpam-Email-Classification_head.safetensors (2.27 MiB); heads/annotationGenHead.safetensors (56.80 MiB); heads/cyPhishing-Email-Detection_head.safetensors (2.28 MiB); heads/eaphishing-dataset_head.safetensors (2.27 MiB); +3 more model artifacts

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Newer 58.3M encoder with seven separate Safetensors heads, including four-class malicious URL and binary phishing heads. No standard sequence-classification head is wired in config; the Hub example defines custom PyTorch modules.	<code>tokenizer_class=DistilBertTokenizer; model_max_length=1000000000000000019884624838656</code>	Head-specific; four-class head 0 benign, 1 defacement, 2 phishing, 3 malware

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	CPU/GPU custom worker	FP32 222.32 MiB; FP16/BF16 111.16 MiB; INT8 55.58 MiB; INT4 27.79 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Card reports task-head results including 0.9982 accuracy for a four-class head; heads require custom code and independent reproduction.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: BENCHMARK AFTER V4. This repository is **VERIFIED - REQUIRES CUSTOM CODE** and remains no higher than R1 from public evidence.

Link repository dossier 4/9

elftsdmr/malware-url-detect

Decision	Acquisition	Readiness	Repository status	Current compatibility
REJECT	CLASS E	R0	REJECTED	No provider; input is domain-only

Role	Turkey-focused URL classifier
Repository	elftsdmr/malware-url-detect
Pinned revision	647f1adf5305e90235ce2a741b9c3351c22ae458
Checked / modified	Checked 15 August 2026; Hub lastModified=2023-08-26T11:09:23.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / transformers
Architecture	architectures=['BertForSequenceClassification']; model_type=bert; num_hidden_layers=12; hidden_size=768; num_attention_heads=12; max_position_embeddings=512
Parameters	178M (published/engineering estimate or UNKNOWN)
Repository storage	1.99 GiB
Primary artifacts	pytorch_model.bin (678.53 MiB); training_args.bin (3.56 KiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
The card states an unknown dataset, Turkey-specific harmful links and domain-only input. This is not a qualified multiclass URL model and has weak provenance.	tokenizer_class=BertTokenizer; model_max_length=512	0=BENIGN, 1=MALWARE

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	None	FP32 679.02 MiB; FP16/BF16 339.51 MiB; INT8 169.75 MiB; INT4 84.88 MiB	unsafe weight files=2; repository Python files=0; Safetensors params=not reported

Evidence, licensing and qualification

Card reports 0.945 accuracy on an unknown Turkey-focused dataset; insufficient provenance.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	UNKNOWN / not declared	REJECT PENDING DATA PROVENANCE	Domain/campaign/time-grouped; newly registered and compromised legitimate domains; shorteners; IDN/confusables; malformed/encoded URLs; class-specific phishing/malware/defacement; India lures; false positives per 10k.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: REJECT. This repository is REJECTED and remains no higher than R0 from public evidence.

[Link repository dossier 5/9](#)

microsoft/deberta-v3-small

Decision	Acquisition	Readiness	Repository status	Current compatibility
FINE-TUNE FIRST	CLASS B	R1	VERIFIED - BASE MODEL ONLY	No task head; async worker required

Role	Sanitized page-text fine-tuning backbone
Repository	microsoft/deberta-v3-small
Pinned revision	a36c739020e01763fe789b4b85e2df55d6180012
Checked / modified	Checked 15 August 2026; Hub lastModified=2022-09-26T08:59:13.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	fill-mask / transformers
Architecture	model_type=deberta-v2; num_hidden_layers=6; hidden_size=768; num_attention_heads=12; max_position_embeddings=512
Parameters	142M (published/engineering estimate or UNKNOWN)
Repository storage	3.59 GiB
Primary artifacts	pytorch_model.bin (272.81 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Masked-language-model backbone only. Fine-tune on sanitized title/visible text/form-label outputs; never fetch or execute pages inside the model process.	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files	No task label map - base/embedding model or UNKNOWN

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
hf-inference: live / fill-mask	CPU/modest GPU after tuning	FP32 541.69 MiB; FP16/BF16 270.84 MiB; INT8 135.42 MiB; INT4 67.71 MiB	unsafe weight files=1; repository Python files=0; Safetensors params=not reported

Evidence, licensing and qualification

No directly decision-grade published/card metric was found for this VeriTrust role; benchmark evidence remains insufficient.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Domain/campaign/time-grouped; newly registered and compromised legitimate domains; shorteners; IDN/confusables; malformed/encoded URLs; class-specific phishing/malware/defacement; India lures; false positives per 10k.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: FINE-TUNE FIRST. This repository is VERIFIED - BASE MODEL ONLY and remains no higher than R1 from public evidence.

Link repository dossier 6/9

google/siglip2-base-patch16-224

Decision	Acquisition	Readiness	Repository status	Current compatibility
DOWNLOAD + SELF-HOST	CLASS A	R2	VERIFIED - NO HF PROVIDER	Embedding worker and reference index required

Role	Screenshot/approved-brand embedding model
Repository	google/siglip2-base-patch16-224
Pinned revision	75de2d55ec2d0b4efc50b3e9ad70dba96a7b2fa2
Checked / modified	Checked 15 August 2026; Hub lastModified=2025-02-21T04:03:35.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	zero-shot-image-classification / transformers
Architecture	model_type=siglip
Parameters	375M (exact Safetensors metadata)
Repository storage	2.83 GiB
Primary artifacts	model.safetensors (1.40 GiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Multilingual vision-language encoder for embeddings/zero-shot retrieval. It provides brand-similarity evidence only; no malicious verdict or generic HF raw-image classifier contract is available.	size={'height': 224, 'width': 224}; do_resize=True; do_normalize=True; do_rescale=True; rescale_factor=0.00392156862745098; image_mean=[0.5, 0.5, 0.5]; image_std=[0.5, 0.5, 0.5]; resample=2; image_processor_type=SiglipImageProcessor	No task label map - base/embedding model or UNKNOWN

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	Modest GPU; CPU research	FP32 1.40 GiB; FP16/BF16 715.61 MiB; INT8 357.81 MiB; INT4 178.90 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=yes

Evidence, licensing and qualification

No directly decision-grade published/card metric was found for this VeriTrust role; benchmark evidence remains insufficient.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
apache-2.0	UNKNOWN / not declared	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Domain/campaign/time-grouped; newly registered and compromised legitimate domains; shorteners; IDN/confusables; malformed/encoded URLs; class-specific phishing/malware/defacement; India lures; false positives per 10k.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: DOWNLOAD + SELF-HOST. This repository is VERIFIED - NO HF PROVIDER and remains no higher than R2 from public evidence.

Link repository dossier 7/9

protectai/codebert-base-Malicious_URLs-onnx

Decision	Acquisition	Readiness	Repository status	Current compatibility
REJECT	CLASS E	R0	VERIFIED - ARCHIVED/STALE	Archived and no current provider

Role	CodeBERT ONNX malicious URL model
Repository	protectai/codebert-base-Malicious_URLs-onnx
Pinned revision	6e77d4e5b602736f8ec048463242e41894884844
Checked / modified	Checked 15 August 2026; Hub lastModified=2026-07-09T16:25:46.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / transformers
Architecture	architectures=['RobertaForSequenceClassification']; model_type=roberta; num_hidden_layers=12; hidden_size=768; num_attention_heads=12; max_position_embeddings=514
Parameters	125M (published/engineering estimate or UNKNOWN)
Repository storage	475.77 MiB
Primary artifacts	model.onnx (475.77 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
The card explicitly says the project is archived and unmaintained; model license is absent. A fresh, licensed export from an active source would be required.	tokenizer_class=RobertaTokenizer; model_max_length=512; padding_side=right; truncation_side=right	0=phishing, 1=benign, 2=defacement, 3=malware

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	None	FP32 476.84 MiB; FP16/BF16 238.42 MiB; INT8 119.21 MiB; INT4 59.60 MiB	unsafe weight files=0; repository Python files=0; Safetensors params=not reported

Evidence, licensing and qualification

No directly decision-grade published/card metric was found for this VeriTrust role; benchmark evidence remains insufficient.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
UNKNOWN	['sid321axn/malicious-urls-dataset']	LICENSE UNKNOWN - REJECT	Domain/campaign/time-grouped; newly registered and compromised legitimate domains; shorteners; IDN/confusables; malformed/encoded URLs; class-specific phishing/malware/defacement; India lures; false positives per 10k.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: REJECT. This repository is VERIFIED - ARCHIVED/STALE and remains no higher than R0 from public evidence.

Link repository dossier 8/9

pirocheto/phishing-url-detection

Decision	Acquisition	Readiness	Repository status	Current compatibility
DOWNLOAD ONNX + BENCHMARK	CLASS A	R2	VERIFIED - NO HF PROVIDER	Dedicated ONNX adapter

Role	Independent linear-SVM phishing shadow
Repository	pirocheto/phishing-url-detection
Pinned revision	44f3b19f705b52532e0aadf3d0d15dd892b8a2fb
Checked / modified	Checked 15 August 2026; Hub lastModified=2023-12-01T09:59:20.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / sklearn
Architecture	Not fully declared in standard config; inspect pinned upstream runtime
Parameters	UNKNOWN (published/engineering estimate or UNKNOWN)
Repository storage	1.49 GiB
Primary artifacts	model.onnx (22.46 MiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
Character/text pipeline in ONNX provides stronger architectural diversity from URLBERT. Never load the included pickle in production; acquire only the ONNX artifact and verify feature equivalence.	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files	ONNX probability index 1=phishing; index 0=benign

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	Tiny CPU ONNX worker	UNKNOWN / not meaningful	unsafe weight files=1; repository Python files=0; Safetensors params=not reported

Evidence, licensing and qualification

Card reports ROC-AUC 0.986844 and accuracy 0.948568; use ONNX only and reproduce domain/time-grouped splits.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	['pirocheto/phishing-url']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW	Domain/campaign/time-grouped; newly registered and compromised legitimate domains; shorteners; IDN/confusables; malformed/encoded URLs; class-specific phishing/malware/defacement; India lures; false positives per 10k.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: DOWNLOAD ONNX + BENCHMARK. This repository is VERIFIED - NO HF PROVIDER and remains no higher than R2 from public evidence.

[Link repository dossier 9/9](#)

DiligentAI/urlbert-url-classifier

Decision	Acquisition	Readiness	Repository status	Current compatibility
REJECT	CLASS E	R0	REJECTED	Wrong task and custom code

Role	Official-site vs platform URL classifier
Repository	DiligentAI/urlbert-url-classifier
Pinned revision	e7b707843407d45fdcfec2844517eaf5ef56a63c
Checked / modified	Checked 15 August 2026; Hub lastModified=2026-02-10T00:32:22.000Z
Visibility	public=True; gated=False; disabled=False
Pipeline / library	text-classification / not declared
Architecture	architectures=['BertModel']; model_type=bert; num_hidden_layers=12; hidden_size=768; num_attention_heads=12; max_position_embeddings=512
Parameters	111M (exact Safetensors metadata)
Repository storage	5.77 GiB
Primary artifacts	model.safetensors (422.00 MiB); training_args.bin (5.64 KiB)

Scientific and I/O contract

Actual task / fit	Pinned preprocessing/tokenizer	Output contract
It classifies official websites versus third-party platforms, not benign versus malicious. Repository name similarity is not task fit.	tokenizer_class=PreTrainedTokenizerFast; model_max_length=1024	No task label map - base/embedding model or UNKNOWN

Runtime, provider and memory

Provider check	Recommended path	Raw parameter-memory lower bounds	Supply-chain surface
No live Inference Provider mapping	None	FP32 421.98 MiB; FP16/BF16 210.99 MiB; INT8 105.50 MiB; INT4 52.75 MiB	unsafe weight files=1; repository Python files=1; Safetensors params=yes

Evidence, licensing and qualification

No directly decision-grade published/card metric was found for this VeriTrust role; benchmark evidence remains insufficient.

Model license	Dataset declaration	Commercial status	VeriTrust benchmark
mit	[custom]	REJECT	Domain/campaign/time-grouped; newly registered and compromised legitimate domains; shorteners; IDN/confusables; malformed/encoded URLs; class-specific phishing/malware/defacement; India lures; false positives per 10k.

Calibration gate: raw logits/scores are not probabilities. Preserve raw outputs, fit a versioned calibrator on held-out VeriTrust-like data, report ECE/Brier/reliability, select thresholds from cost/coverage objectives, and require explicit unknown-output handling.

Disposition: REJECT. This repository is REJECTED and remains no higher than R0 from public evidence.

20. Unified Gateway model decision

Decision: do not acquire or search for a generic gateway AI classifier. The blueprint and Unified Gateway handbook recommend deterministic, versioned evidence correlation initially. The executable repository also implements deterministic max/bonus logic rather than a learned meta-model, although its current rule set is provisional.

A learned meta-classifier is deferred until VeriTrust has sufficient decision-quality labels collected under stable upstream evidence contracts, with entity/campaign/tenant/time separation and no upstream-score leakage between train and test. It must demonstrate incremental calibration and selective-risk benefit over the deterministic policy, not merely higher aggregate accuracy.

Initial gateway obligation	Implementation meaning
Immutable input evidence	Model/processor/calibrator/benchmark IDs bound before inference; inference receipt recorded
Expected-evidence ledger	Know which families were requested, returned, abstained, timed out or failed
Versioned deterministic policy	Explicit correlation rules and bounded bonuses; replayable by policy version
No silent substitution	Do not replace a missing scientific family with an alias fallback or local heuristic
Selective action	High-impact actions require completeness, calibration and human-review paths
Future meta-model gate	Leakage-safe dataset, baseline comparison, calibration, explainability, rollback and shadow run

21. Learning Module model decision

Decision: no core Hugging Face inference model is required. The Learning Module handbook defines content, courses, assessments, progress, certification, analytics and administration. It does not justify manufacturing an AI model dependency.

Optional future uses - translation drafts, accessibility summaries, question suggestions or bounded content tagging - must remain assistive, opt-in, reviewable and separately governed. An LLM must never grade authoritative certification answers, generate unreviewed security truth or become necessary for basic course operation.

22. Parameter-range guidance

Parameter count is a capacity/cost constraint, not a quality rank. The ranges below are engineering starting points for VeriTrust benchmark design. Data coverage, leakage control, preprocessing and calibration frequently dominate additional parameters.

Role	Minimum sensible	Recommended practical	Upper worth evaluating	Diminishing returns	Why
Face detection / landmarks	0.1-5M	0.5-2M	5-15M	Above ~15M	Geometry/preprocessing; deterministic speed and recall matter more than semantic capacity.
Facial manipulation	20-80M	80-450M	450-900M	Above ~450M without new data	Cross-manipulation generalization and aligned-face preprocessing dominate; ViT-L is justified only if it wins held-out tests.
Synthetic full-frame image	15-30M	20-100M	100-400M	Above ~100M	Small ViTs can capture generator artifacts; data diversity is more valuable than a larger backbone.
Frequency/residual forensic	1-15M	5-40M	40-100M	Above ~40M	The goal is an independent low-level evidence family, not semantic capacity.
Phishing fast text	20-40M	50-100M	100-200M	Above ~100M	CPU latency, truncation and campaign-split generalization make DistilBERT-class models practical.
Phishing multilingual	80-150M	150-300M	300-500M	Above ~300M	Tokenizer and locale coverage justify more capacity; labels and data coverage remain limiting.
Phishing shadow	100-200M	200-450M	450-900M	Above ~450M	Shadow capacity is useful only if it produces genuinely different errors and survives real-world holdouts.
Bounded semantic extractor	1-3B	3-8B	8-14B	Above ~8B	Small instruct models can emit bounded JSON; larger models add cost and privacy surface without truth authority.
URL multiclass	1-4M	3-30M	30-100M	Above ~60M	URLs are short and structured; tokenizer/data quality and temporal splits dominate parameter count.
URL binary shadow	0.1-4M	1-30M	30-100M	Above ~30M	Prefer a different feature family over another larger URL Transformer.
Sanitized page text	20-60M	60-150M	150-300M	Above ~150M	Async evidence benefits from context but should stay cheap and isolated from rendering.
Screenshot/brand embedding	80-150M	200-400M	400-900M	Above ~400M	Reference retrieval quality and image preprocessing matter; base SigLIP2 is already medium-heavy.

For GBDT/linear models, parameter count is not comparable with Transformer parameters. Report tree count/depth/features or vectorizer vocabulary/non-zero coefficients separately.

23. Disk-size and RAM/VRAM matrix

Repository	Params	Hub repo size	Raw parameter lower bounds	Dev RAM	Server RAM	GPU VRAM guidance	Benchmark batches CPU/GPU	Class
OwensLab/commfor-model-384	21.8M	83.22 MiB	FP32 83.21 MiB; FP16/BF16 41.60 MiB; INT8 20.80 MiB; INT4 10.40 MiB	8 GB	8-16 GB	Optional 4-8 GB	1/8/32	lightweight
OwensLab/commfor-model-224	21.7M	82.66 MiB	FP32 82.65 MiB; FP16/BF16 41.32 MiB; INT8 20.66 MiB; INT4 10.33 MiB	8 GB	8-16 GB	Optional 4-8 GB	1/8/32	lightweight
yemandy/deepfake-detection	UNKNOWN	1.70 GiB	UNKNOWN / not meaningful	16-32 GB	32 GB	8-16 GB	1/4/8	medium-heavy
openai/clip-vit-large-patch14	428M	12.76 GiB	FP32 1.59 GiB; FP16/BF16 815.61 MiB; INT8 407.81 MiB; INT4 203.90 MiB	16-32 GB	32 GB	8-16 GB	1/4/8	medium-heavy
buildborderless/CommunityForensics-DeepfakeDet-VIT	21.8M	1.26 GiB	FP32 83.21 MiB; FP16/BF16 41.60 MiB; INT8 20.80 MiB; INT4 10.40 MiB	8 GB	8-16 GB	Optional 4-8 GB	1/8/32	lightweight
dima806/deepfake_vs_real_image_detection	85.8M	3.84 GiB	FP32 327.30 MiB; FP16/BF16 163.65 MiB; INT8 81.83 MiB; INT4 40.91 MiB	8 GB	8-16 GB	Optional 4-8 GB	1/8/32	lightweight
capcheck/ai-image-detection	85.8M	327.32 MiB	FP32 327.30 MiB; FP16/BF16 163.65 MiB; INT8 81.83 MiB; INT4 40.91 MiB	8 GB	8-16 GB	Optional 4-8 GB	1/8/32	lightweight
cybersectory/phishing-email-detection-distilbert_v2.4.1	67M	255.43 MiB	FP32 255.42 MiB; FP16/BF16 127.71 MiB; INT8 63.85 MiB; INT4 31.93 MiB	8 GB	8-16 GB	Optional 4-8 GB	1/8/32	lightweight
dima806/phishing-email-detection	65.8M	2.45 GiB	FP32 250.94 MiB; FP16/BF16 125.47 MiB; INT8 62.74 MiB; INT4 31.37 MiB	8 GB	8-16 GB	Optional 4-8 GB	1/8/32	lightweight
ealvaradob/bert-finetuned-phishing	336M	39.96 GiB	FP32 1.25 GiB; FP16/BF16 640.87 MiB; INT8 320.43 MiB; INT4 160.22 MiB	16-32 GB	32 GB	8-16 GB	1/4/8	medium-heavy
microsoft/mdeberta-v3-base	279M	5.45 GiB	FP32 1.04 GiB; FP16/BF16 532.15 MiB; INT8 266.08 MiB; INT4 133.04 MiB	16-32 GB	32 GB	8-16 GB	1/4/8	medium-heavy
mistralai/Mistral-7B-Instruct-v0.3	7.25B	27.00 GiB	FP32 27.00 GiB; FP16/BF16 13.50 GiB; INT8 6.75 GiB; INT4 3.38 GiB	64 GB	64-128 GB	16-24+ GB	1/1/2	heavy
FacebookAI/xlm-roberta-base	279M	5.92 GiB	FP32 1.04 GiB; FP16/BF16 531.93 MiB; INT8 265.97 MiB; INT4 132.98 MiB	16-32 GB	32 GB	8-16 GB	1/4/8	medium-heavy
Sonje03/phishlens-distilbert	67M	255.43 MiB	FP32 255.41 MiB; FP16/BF16 127.71 MiB; INT8 63.85 MiB; INT4 31.93 MiB	8 GB	8-16 GB	Optional 4-8 GB	1/8/32	lightweight
mikaenurminen/modernbert-large-phishing	396M	4.20 GiB	FP32 1.47 GiB; FP16/BF16 754.99 MiB; INT8 377.50 MiB; INT4 188.75 MiB	16-32 GB	32 GB	8-16 GB	1/4/8	medium-heavy
CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	3.69M	14.08 MiB	FP32 14.06 MiB; FP16/BF16 7.03 MiB; INT8 3.52 MiB; INT4 1.76 MiB	4 GB	4-8 GB	Not needed	1/16/64	tiny
CrabInHoney/urlbert-tiny-v4-phishing-classifier	3.69M	14.08 MiB	FP32 14.06 MiB; FP16/BF16 7.03 MiB; INT8 3.52 MiB; INT4 1.76 MiB	4 GB	4-8 GB	Not needed	1/16/64	tiny
CrabInHoney/urlbert-tiny-v5	58.3M	292.77 MiB	FP32 222.32 MiB; FP16/BF16 111.16 MiB; INT8 55.58 MiB; INT4 27.79 MiB	8 GB	8-16 GB	Optional 4-8 GB	1/8/32	lightweight
microsoft/deberta-v3-small	142M	3.59 GiB	FP32 541.69 MiB; FP16/BF16 270.84 MiB; INT8 135.42 MiB; INT4 67.71 MiB	16 GB	16-32 GB	Optional 8 GB	1/4/16	medium
google/siglip2-base-patch16-224	375M	2.83 GiB	FP32 1.40 GiB; FP16/BF16 715.61 MiB; INT8 357.81 MiB; INT4 178.90 MiB	16-32 GB	32 GB	8-16 GB	1/4/8	medium-heavy
pirocheto/phishing-url-detection	UNKNOWN	1.49 GiB	UNKNOWN / not meaningful	4 GB	4-8 GB	Not needed	1/16/64	tiny

Actual runtime RAM/VRAM is higher than raw parameter memory because of activations, framework/runtime overhead, tokenizers/processors, CUDA contexts, batching, temporary tensors and - for causal LLMs - KV cache. All latency and throughput are BENCHMARK REQUIRED; no invented milliseconds are used.

24. Inference/runtime matrix and normalized scorecard

Repository	Role	Real inference path	HF mapping on 2026-08-15	Current adapter	Best profile
OwensLab/commfor-model-384	Full-frame synthetic-image primary	SELF-HOSTED GPU/CUSTOM CODE	No live Inference Provider mapping	Dedicated worker only	Modest GPU; CPU benchmark
OwensLab/commfor-model-224	Full-frame synthetic-image latency alternative	SELF-HOSTED GPU/CUSTOM CODE	No live Inference Provider mapping	Dedicated worker only	CPU/modest GPU benchmark
yermandy/deepfake-detection	Facial manipulation detector	SELF-HOSTED GPU/CUSTOM CODE	No live Inference Provider mapping	Preprocessed aligned-face worker only	Production GPU worker
openai/clip-vit-large-patch14	Facial-forensics backbone / embedding baseline	TRAIN/FINE-TUNE BEFORE INFERENCE	No live Inference Provider mapping	Not a detector; processor/worker required	16-24 GB GPU for tuning; GPU inference
buildborderless/CommunityForensics-DeepfakeDet-ViT	Synthetic-image standard-Transformers challenger	SELF-HOSTED TRANSFORMERS / ENDPOINT	hf-inference: error / image-classification	Adapter refactor; single-logit sigmoid	CPU ONNX / modest GPU
dima806/deepfake_vs_real_image_detection	Legacy real-vs-fake image fallback	DIRECT HF SERVERLESS (current mapping)	hf-inference: live / image-classification	Configuration-only for current raw-image adapter	HF serverless currently live; CPU/GPU
capcheck/ai-image-detection	CIFAKE synthetic-image fallback	SELF-HOSTED TRANSFORMERS / ENDPOINT	No live Inference Provider mapping	Endpoint/worker refactor	Self-hosted Transformers / ONNX export
cybersectory/phishing-email-detection-distilbert_v2.4.1	Mixed email/URL four-class compatibility candidate	DIRECT HF SERVERLESS (current mapping)	hf-inference: live / text-classification	Current MailGuard label collapse matches card mapping	HF serverless currently live; CPU
dima806/phishing-email-detection	Binary DistilBERT fallback	SELF-HOSTED TRANSFORMERS / ENDPOINT	No live Inference Provider mapping	Label-map and endpoint refactor	Self-hosted CPU
ealvaradob/bert-finetuned-phishing	Large English shadow classifier	DIRECT HF SERVERLESS (current mapping)	hf-inference: live / text-classification	MailGuard benign-label fix required	HF serverless / GPU or strong CPU
microsoft/mdeberta-v3-base	Preferred multilingual fine-tuning backbone	TRAIN/FINE-TUNE BEFORE INFERENCE	hf-inference: live / fill-mask	No classification head; new worker	Modest/production GPU for tuning; CPU/GPU inference
mistralai/Mistral-7B-Instruct-v0.3	Optional bounded observation extractor	SELF-HOSTED TRANSFORMERS / ENDPOINT	novita: error / conversational	Chat adapter shape fits, but provider is error and prompt role is wrong	16-24 GB GPU; self-host
FacebookAI/xlm-roberta-base	Multilingual fine-tuning fallback	TRAIN/FINE-TUNE BEFORE INFERENCE	hf-inference: live / fill-mask	No classification head; new worker	Modest GPU tuning; CPU/GPU inference
Sonje03/phishlens-distilbert	Preferred fast English classifier candidate	SELF-HOSTED TRANSFORMERS / ENDPOINT	No live Inference Provider mapping	Label-map and self-host worker required	Standard CPU worker
mikaelnurminen/modernbert-large-phishing	Large shadow/error-diversity classifier	SELF-HOSTED TRANSFORMERS / ENDPOINT	No live Inference Provider mapping	Dedicated worker; never current adapter	Modest GPU; eager attention
CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	Preferred fast four-class URL candidate	SELF-HOSTED TRANSFORMERS / ENDPOINT	No live Inference Provider mapping	Label-map swap and self-host worker required	Standard CPU worker / ONNX export
CrabInHoney/urlbert-tiny-v4-phishing-classifier	Binary phishing URL shadow	SELF-HOSTED TRANSFORMERS / ENDPOINT	No live Inference Provider mapping	Self-host worker; explicit label map	Standard CPU worker / ONNX export
CrabInHoney/urlbert-tiny-v5	New URL encoder + task-head challenger	SELF-HOSTED CUSTOM CODE	No live Inference Provider mapping	Custom head loader required	CPU/GPU custom worker
microsoft/deberta-v3-small	Sanitized page-text fine-tuning backbone	TRAIN/FINE-TUNE BEFORE INFERENCE	hf-inference: live / fill-mask	No task head; async worker required	CPU/modest GPU after tuning
google/siglip2-base-patch16-224	Screenshot/approved-brand embedding model	SELF-HOSTED TRANSFORMERS / ENDPOINT	No live Inference Provider mapping	Embedding worker and reference index required	Modest GPU; CPU research
pirocheto/phishing-url-detection	Independent linear-SVM phishing shadow	SELF-HOSTED ONNX	No live Inference Provider mapping	Dedicated ONNX adapter	Tiny CPU ONNX worker

Scorecard block A - scientific/governance (0=poor, 5=strong; N/A excluded)

Repository	Task fit	Provenance	License clarity	Model card	Artifact safety	Error diversity
OwensLab/commfor-model-384	5	5	5	3	5	4
OwensLab/commfor-model-224	5	5	5	3	5	1
yemandy/deepfake-detection	5	5	5	4	1	4
openai/clip-vit-large-patch14	3	5	0	4	3	2
buildborderless/CommunityForensics-DeepfakeDet-ViT	5	3	5	5	3	2
dima806/deepfake_vs_real_image_detection	3	3	5	3	3	2
capcheck/ai-image-detection	3	3	5	4	5	1
cybersectony/phishing-email-detection-distilbert_v2.4.1	3	3	5	3	5	2
dima806/phishing-email-detection	4	2	5	1	3	1
ealvaradob/bert-finetuned-phishing	4	4	5	4	1	3
microsoft/mdeberta-v3-base	4	5	5	4	1	4
mistralai/Mistral-7B-Instruct-v0.3	3	5	5	5	5	4
FacebookAI/xlm-roberta-base	3	5	5	5	3	3
Sonje03/phishlens-distilbert	5	4	5	5	5	3
mikaelnurminen/modernbert-large-phishing	5	4	5	5	5	4
CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	5	4	5	4	5	3
CrabInHoney/urlbert-tiny-v4-phishing-classifier	5	4	5	4	5	1
CrabInHoney/urlbert-tiny-v5	5	4	5	5	5	2
microsoft/deberta-v3-small	4	5	5	4	1	4
google/siglip2-base-patch16-224	5	5	5	5	5	5
pirocheto/phishing-url-detection	4	4	5	5	1	5

Scorecard block B - efficiency/contract (0=poor, 5=strong)

Repository	CPU efficiency	GPU efficiency	Parameter efficiency	Storage footprint	Preprocessing clarity	Label clarity
OwensLab/commfor-model-384	4	5	4	4	4	4
OwensLab/commfor-model-224	4	5	4	4	4	4
yemandy/deepfake-detection	1	2	1	3	3	2
openai/clip-vit-large-patch14	2	3	2	1	5	0
buildborderless/CommunityForensics-DeepfakeDet-ViT	4	5	4	3	5	5
dima806/deepfake_vs_real_image_detection	4	5	4	2	5	5
capcheck/ai-image-detection	4	5	4	4	5	5
cybersectony/phishing-email-detection-distilbert_v2.4.1	4	5	4	4	4	4
dima806/phishing-email-detection	4	5	4	2	2	4
ealvaradob/bert-finetuned-phishing	2	3	2	0	4	5
microsoft/mdeberta-v3-base	2	3	2	2	4	0
mistralai/Mistral-7B-Instruct-v0.3	0	1	0	0	5	0
FacebookAI/xlm-roberta-base	2	3	2	2	4	0
Sonje03/phishlens-distilbert	4	5	4	4	5	5
mikaelnurminen/modernbert-large-phishing	2	3	2	2	5	5
CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	5	5	5	5	5	5
CrabInHoney/urlbert-tiny-v4-phishing-classifier	5	5	5	5	5	4
CrabInHoney/urlbert-tiny-v5	4	5	4	4	4	5
microsoft/deberta-v3-small	3	4	3	2	4	0
google/siglip2-base-patch16-224	2	3	2	2	5	0
pirocheto/phishing-url-detection	5	5	5	3	5	5

Scorecard block C - deployment/readiness (0=poor, 5=strong)

Repository	Inference compatibility	Multilingual fit	Benchmark evidence	Calibration readiness	HF deployment	Implementation simplicity
OwensLab/commfor-model-384	3	N/A	4	2	2	2
OwensLab/commfor-model-224	3	N/A	4	2	2	2
yemandy/deepfake-detection	3	N/A	4	1	2	1
openai/clip-vit-large-patch14	3	N/A	5	1	2	3
buildborderless/CommunityForensics-DeepfakeDet-ViT	2	N/A	4	2	0	3
dima806/deepfake_vs_real_image_detection	5	N/A	2	2	5	5
capcheck/ai-image-detection	3	N/A	2	2	2	4
cybersectory/phishing-email-detection-distilbert_v2.4.1	2	1	2	2	5	5
dima806/phishing-email-detection	2	1	1	2	2	3
ealvaradob/bert-finetuned-phishing	2	1	3	2	5	3
microsoft/mdeberta-v3-base	3	5	5	1	5	3
mistralai/Mistral-7B-Instruct-v0.3	2	4	5	1	0	1
FacebookAI/xlm-roberta-base	3	5	5	1	5	3
Sonje03/phishlens-distilbert	3	1	3	2	2	4
mikaelnurminen/modernbert-large-phishing	3	1	4	3	2	2
CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	3	N/A	3	2	2	5
CrabInHoney/urlbert-tiny-v4-phishing-classifier	3	N/A	3	2	2	5
CrabInHoney/urlbert-tiny-v5	2	N/A	3	2	2	1
microsoft/deberta-v3-small	3	N/A	5	1	5	3
google/siglip2-base-patch16-224	3	N/A	5	1	2	3
pirocheto/phishing-url-detection	2	N/A	3	2	2	5

Critical overrides: non-commercial or unknown licensing, wrong task, unverified labels, stale/archived artifacts and provenance conflicts cause rejection regardless of any numeric average. Scores are a comparison aid, not a qualification threshold.

25. HF-provider and self-hosting matrix

Repository	Hub task	Live provider mapping	Fine-tune first	Custom repository code	Recommended deployment
OwensLab/commfor-model-384	none	No live Inference Provider mapping	No	No/none in repo	Modest GPU; CPU benchmark
OwensLab/commfor-model-224	none	No live Inference Provider mapping	No	No/none in repo	CPU/modest GPU benchmark
yemandy/deepfake-detection	image-classification	No live Inference Provider mapping	No	No/none in repo	Production GPU worker
openai/clip-vit-large-patch14	zero-shot-image-classification	No live Inference Provider mapping	Yes	No/none in repo	16-24 GB GPU for tuning; GPU inference

Repository	Hub task	Live provider mapping	Fine-tune first	Custom repository code	Recommended deployment
buildborderless/CommunityForensics-DeepfakeDet-ViT	image-classification	hf-inference: error / image-classification	No	Yes	CPU ONNX / modest GPU
onnx-community/CommunityForensics-DeepfakeDet-ViT-ONNX	image-classification	No live Inference Provider mapping	No	No/none in repo	None
dima806/deepfake_vs_real_image_detection	image-classification	hf-inference: live / image-classification	No	No/none in repo	HF serverless currently live; CPU/GPU
capcheck/ai-image-detection	image-classification	No live Inference Provider mapping	No	No/none in repo	Self-hosted Transformers / ONNX export
prithivMLmods/AI-vs-Deepfake-vs-Real	image-classification	No live Inference Provider mapping	No	No/none in repo	None
cybersectory/phishing-email-detection-distilbert_v2.4.1	text-classification	hf-inference: live / text-classification	No	No/none in repo	HF serverless currently live; CPU
dima806/phishing-email-detection	text-classification	No live Inference Provider mapping	No	No/none in repo	Self-hosted CPU
aamoshdahal/email-phishing-distilbert-finetuned	text-classification	No live Inference Provider mapping	No	No/none in repo	None
ealvaradob/bert-finetuned-phishing	text-classification	hf-inference: live / text-classification	No	No/none in repo	HF serverless / GPU or strong CPU
microsoft/mdeberta-v3-base	fill-mask	hf-inference: live / fill-mask	Yes	No/none in repo	Modest/production GPU for tuning; CPU/GPU inference
mistralai/Mistral-7B-Instruct-v0.3	none	novita: error / conversational	No	No/none in repo	16-24 GB GPU; self-host
FacebookAI/xlm-roberta-base	fill-mask	hf-inference: live / fill-mask	Yes	No/none in repo	Modest GPU tuning; CPU/GPU inference
Sonje03/phishlens-distilbert	text-classification	No live Inference Provider mapping	No	No/none in repo	Standard CPU worker
mikaelnurminen/modernbert-large-phishing	text-classification	No live Inference Provider mapping	No	No/none in repo	Modest GPU; eager attention
JustAPR/email-phishing-distilbert-finetuned	text-classification	No live Inference Provider mapping	No	No/none in repo	None
Qwen/Qwen2.5-3B-Instruct	text-generation	featherless-ai: live / conversational	No	No/none in repo	None for commercial product
CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	text-classification	No live Inference Provider mapping	No	No/none in repo	Standard CPU worker / ONNX export
CrabInHoney/urlbert-tiny-v4-phishing-classifier	text-classification	No live Inference Provider mapping	No	No/none in repo	Standard CPU worker / ONNX export
CrabInHoney/urlbert-tiny-v5	feature-extraction	No live Inference Provider mapping	No	No/none in repo	CPU/GPU custom worker
elftsdmr/malware-url-detect	text-classification	No live Inference Provider mapping	No	No/none in repo	None
microsoft/deberta-v3-small	fill-mask	hf-inference: live / fill-mask	Yes	No/none in repo	CPU/modest GPU after tuning
google/siglip2-base-patch16-224	zero-shot-image-classification	No live Inference Provider mapping	No	No/none in repo	Modest GPU; CPU research
protectai/codebert-base-Malicious_URLs-onnx	text-classification	No live Inference Provider mapping	No	No/none in repo	None
pirocheto/phishing-url-detection	text-classification	No live Inference Provider mapping	No	No/none in repo	Tiny CPU ONNX worker
DiligentAI/urlbert-url-classifier	text-classification	No live Inference Provider mapping	No	Yes	None

A model page is not evidence of serverless availability. Provider status was queried through the official Hub API `expand=inferenceProviderMapping`. Dedicated Inference Endpoints remain technically possible for many standard models, but VeriTrust must build/pin the container and endpoint URL; today's adapter constructs the generic HF router URL and cannot target an arbitrary endpoint.

26. Preprocessing and input contracts

Role	Required preprocessing	Tensor/input	Qualification warning
Deepfake face manipulation	Canonical RGB -> YuNet boxes/5 landmarks -> DeepfakeBench-compatible alignment/crop -> model-specific CLIP normalization	Face child artifact, tensor NCHW	Pin detector, alignment geometry, crop size, interpolation, CLIP mean/std
Community Forensics 384	RGB; official pinned upstream transforms. Third-party corrected integration: shortest edge 440, center crop 384, CLIP normalize	Full-frame child artifact	Do not assume third-party processor equals official checkpoint; golden-tensor parity required
Community Forensics 224	Official pinned upstream 224 recipe	Full-frame child artifact	Benchmark crop/resolution; not independent from 384
ViT CIFAKE fallbacks	RGB 224x224; rescale 1/255; mean/std 0.5	Full frame	Different normalization from CLIP/Community Forensics
PhishLens DistilBERT	English tokenizer; card inference uses max_length=256; subject/body framing must be qualified	input_ids + attention_mask	Pin tokenizer files, truncation and aggregation
ModernBERT shadow	Tokenizer max 8192; card example max 2048; eager attention; batch=1 pending stability reproduction	input_ids + attention_mask	Long-context cost and quoted threads require slices
mDeBERTa/XLM-R multilingual	Base tokenizers; max ~512; VeriTrust framing to be learned during fine-tune	input_ids + attention mask	Locale/Unicode/tokenization data required
URLBERT-v4	Model tokenizer; raw URL string; max_position_embeddings=64	input_ids + token types/attention	Store original+model string; test IDN/encoding/truncation
URLBERT-v5	Custom DistilBERT tokenizer; max 512; [CLS] embedding -> head	Encoder embedding + selected custom head	Pin exact head file and code; do not load every head as independent evidence
pirocheto ONNX	String tensor URL; embedded vectorizer/LinearSVM pipeline	ONNX string input	Never load model.pkl; validate ONNX feature behavior
DeBERTa page text	Sanitized title + visible text + form labels; no script/HTML secrets; chunking learned in fine-tune	Text chunks + aggregation	Collector/normalizer version is part of model version
SigLIP2	RGB 224x224; rescale 1/255; mean/std 0.5; SiglipProcessor	Normalized image embedding	Reference screenshots use identical processor and crop policy

Preprocessing is part of the scientific model. Any change to decode library, EXIF orientation, resize, interpolation, crop, alignment, normalization, tokenizer, Unicode policy, truncation, chunking or aggregation must create a new version and trigger parity/benchmark/calibration gates.

27. Output and label contracts

Model role	Raw output	Meaning	Invalid/unknown behavior
Facial manipulation	Upstream binary logit/probability - exact direction requires reproduction	Real vs manipulated face	Fail NOT IMPLEMENTATION READY until golden real/manipulated fixtures confirm sign/order
Community Forensics official	Upstream binary score; real=0/fake=1 in project data convention	Full-frame generated evidence	Bind wrapper output explicitly; no label-string guessing
buildborderless integration	Single logit	$\text{sigmoid}(\text{logit})=\text{fake}$; $1-\text{p}=\text{real}$	Current raw-image normalizer would likely invert/guess LABEL_0
PhishLens	Two logits -> softmax	0 Safe, 1 Phishing	Current MailGuard must recognize Safe; otherwise phishing can normalize to 1
ModernBERT phishing	Two logits -> softmax	0 benign, 1 phishing	Preserve logits; eager attention; do not inherit card thresholds
Cybersectony v2.4.1	Four logits -> softmax	0 legitimate_email, 1 phishing_url, 2 legitimate_url, 3 phishing_url_alt	Semantics are unusual/mixed; current collapse matches card but requires independent label fixture
URLBERT-v4 multiclass	Four logits -> softmax	0 benign, 1 defacement, 2 malware, 3 phishing	Correct current Swift map before any benchmark
URLBERT-v4 binary	Two logits -> softmax	0 good/safe, 1 phishing	Use explicit registry map; card wording 'fish' must not leak into API
URLBERT-v5 four-class head	Custom head logits -> softmax	0 benign, 1 defacement, 2 phishing, 3 malware	Different order from v4; head identity must be part of key
pirocheto ONNX	ONNX class/probability outputs	index 1 phishing, index 0 benign	Verify output node names/types and probability calibration
SigLIP2	Embedding vector / similarity	No malicious label	Similarity is raw evidence; threshold/calibration is tenant/reference-specific
Mistral extractor	Strict JSON observations + evidence spans	No truth/risk/action label	Schema validation; reject extra fields and prompt-injected instructions

Production rule: if labels cannot be verified from pinned configuration plus executable golden fixtures, the model is NOT IMPLEMENTATION READY. Unknown labels, missing classes, NaN/Inf, non-finite logits, malformed JSON or unexpected tensor shapes produce typed failure/abstention; they are never guessed.

28. Quantization and ONNX analysis

Candidate	Artifacts	Preferred baseline	Quantization	Parity gate
Community Forensics 384	Safetensors official; third-party ONNX variants	FP32 reference; FP16 GPU	INT8 only after generator/OOD parity; Q4 not recommended initially	Fresh export from pinned official checkpoint; compare logits/tensors
yermandy	TorchScript + project runtime; Hub unsafe artifact surface	FP16 GPU benchmark	INT8/INT4 research only	Reproduce precise path first; trace/export in isolated conversion
PhishLens DistilBERT	Safetensors	FP32 CPU baseline; FP16 GPU	Dynamic INT8 sensible on CPU	Export standard sequence classifier; token/logit parity
ModernBERT-large	Safetensors	BF16/FP16 GPU; FP32 baseline	INT8 potentially useful; INT4 generally unjustified for classifier	Eager-attention and NaN tests before/after export
mDeBERTa/XLM-R	Base artifacts; tune first	BF16/FP16 training/inference	INT8 after fine-tuned calibration	Export only final pinned classifier; SentencePiece/tokenizer parity
URLBERT-v4 models	Safetensors	FP32 CPU baseline	Dynamic INT8 highly plausible	ONNX export; all four class logits, long/IDN/malformed URL parity
URLBERT-v5	Safetensors encoder + heads	FP32/BF16 baseline	INT8 only after custom-head parity	Bundle selected head and code into one immutable worker artifact
pirocheto	ONNX available; pickle also present	Use ONNX CPU	Already compact; further quantization may be irrelevant	Never deserialize pickle; verify ONNX ops support
DeBERTa page text	Tune first	FP32/BF16 baseline	Dynamic/static INT8 after calibration	Chunk aggregation parity is part of export
SigLIP2 base	Safetensors	FP16/BF16 GPU	INT8 benchmark; INT4 not initial	Cosine similarity/ranking parity over brand reference set
Mistral 7B	Safetensors shards	BF16/FP16; 8-bit/4-bit lab	INT4 meaningful for lab self-host but JSON quality parity mandatory	Constrained decoding/schema behavior and privacy load tests

No quantized artifact is approved merely because it loads. Required parity includes logits/probabilities, class ordering, decision agreement, calibration drift, difficult slices, OOD behavior and numerical stability. TensorRT is optional only after a stable ONNX/FP reference and demonstrated throughput need.

29. Licensing and commercial-use matrix

Repository	Model license	Declared datasets	Assessment
OwensLab/commfor-model-384	mit	['OwensLab/CommunityForensics', 'OwensLab/CommunityForensics-Small']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
OwensLab/commfor-model-224	mit	['OwensLab/CommunityForensics', 'OwensLab/CommunityForensics-Small']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
yemandy/deepfake-detection	mit	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
openai/clip-vit-large-patch14	UNKNOWN	UNKNOWN	LICENSE UNKNOWN ON MODEL CARD - LEGAL REVIEW
buildborderless/CommunityForensics-DeepfakeDet-ViT	mit	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
onnx-community/CommunityForensics-DeepfakeDet-ViT-ONNX	mit	UNKNOWN	REJECT
dima806/deepfake_vs_real_image_detection	apache-2.0	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
capcheck/ai-image-detection	apache-2.0	['CIFAKE']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
prithivMLmods/AI-vs-Deepfake-vs-Real	apache-2.0	['prithivMLmods/AI-vs-Deepfake-vs-Real']	REJECT PENDING DATA PROVENANCE
cybersectory/phishing-email-detection-distilbert_v2.4.1	apache-2.0	['cybersectory/PhishingEmailDetectionv2.0']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
dima806/phishing-email-detection	apache-2.0	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
aamoshdahal/email-phishing-distilbert-finetuned	UNKNOWN	UNKNOWN	LICENSE UNKNOWN - REJECT
ealvaradob/bert-finetuned-phishing	apache-2.0	['ealvaradob/phishing-dataset']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
microsoft/mdeberta-v3-base	mit	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
mistralai/Mistral-7B-Instruct-v0.3	apache-2.0	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
FacebookAI/xlm-roberta-base	mit	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
Sonje03/phishlens-distilbert	mit	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
mikaelnurminen/modernbert-large-phishing	apache-2.0	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
JustAPR/email-phishing-distilbert-finetuned	UNKNOWN	UNKNOWN	LICENSE UNKNOWN - REJECT
Qwen/Qwen2.5-3B-Instruct	other	UNKNOWN	NON-COMMERCIAL - REJECT
CrabInHoney/urllibert-tiny-v4-malicious-url-classifier	apache-2.0	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
CrabInHoney/urllibert-tiny-v4-phishing-classifier	apache-2.0	['ealvaradob/phishing-dataset']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
CrabInHoney/urllibert-tiny-v5	apache-2.0	['ealvaradob/phishing-dataset', 'cybersectory/PhishingEmailDetectionv2.0', 'zefang-liu/phishing-email-dataset', 'kmack/Phishing_urls']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
elftsdmr/malware-url-detect	apache-2.0	UNKNOWN	REJECT PENDING DATA PROVENANCE
microsoft/deberta-v3-small	mit	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
google/siglip2-base-patch16-224	apache-2.0	UNKNOWN	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
protectai/codebert-base-Malicious_URLs-onnx	UNKNOWN	['sid321axn/malicious-urls-dataset']	LICENSE UNKNOWN - REJECT
pirocheto/phishing-url-detection	mit	['pirocheto/phishing-url']	LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW
DiligentAI/urllibert-url-classifier	mit	['custom']	REJECT

Required legal record for every row: archive the model card and license at the pinned SHA; identify every upstream dataset and redistribution obligation. Model-file permissiveness does not cure dataset uncertainty. MIT/Apache model artifacts are CLEAR FOR EVALUATION but only LIKELY COMMERCIAL-COMPATIBLE - LEGAL REVIEW for product use until upstream datasets, collection rights, biometric/privacy implications and redistribution terms are reviewed. Qwen2.5-3B is non-commercial under its repository license and is rejected absent a separate license.

30. Model supply-chain and security matrix

Control	Observed risk	Required acquisition control
Identity	Mutable owner/repo or main revision	Pin exact SHA; deny mutable production resolution
Integrity	Hub LFS/Xet OIDs are repository metadata, not a local acquisition log	Record SHA-256 for every downloaded artifact and container image
Format	Some repositories contain .bin/.pt/.pth/pickle/legacy weights	Prefer Safetensors/ONNX; isolate and scan any necessary conversion; never deserialize arbitrary pickle in production
Code	Custom model mixins/Python may execute during load	No <code>trust_remote_code</code> in production; vendor/review minimal runtime into a pinned worker
Provenance	Third-party conversions can drift from official checkpoints	Prove tensor/logit equivalence; record source SHA, conversion tool/version and command
Registry	Current model ID can be stored separately from actual provider execution	Bind registry selection, processor, calibrator and evidence receipt before inference
Dependencies	Product has no Python model-worker lock/SBOM	Build per-worker lockfile/SBOM; scan OS, Python, CUDA/ORT and model artifacts
Distribution	Public Hub availability can change	Mirror approved artifacts privately with original license/card/metadata archive
Secrets/PII	HF/chat providers receive user content	Data-flow review, regional/privacy controls, minimization, encryption, retention and provider contracts
Rollback	Aliases resolve mutable configuration	Canary/shadow with immutable artifact keys and fast rollback to prior qualified bundle

Candidate-specific red flags

- yermandy includes a TorchScript/project-specific path: reproduce in an isolated lab, inspect upstream code and create a reviewed internal runtime.
- buildborderless contains archived wrong/legacy artifacts and repository Python: whitelist only exact corrected files; never acquire the whole repo blindly.
- ealvaradob repositories contain extensive redundant checkpoints/unsafe binaries: download only the pinned minimum artifact set.
- pirocheto includes pickle and ONNX: use ONNX only; the card itself warns about pickle code execution.
- Qwen has a license gate, not a technical safety issue alone; do not let live Featherless availability bypass licensing.

31. VeriTrust benchmark requirements

Model-card accuracy is discovery evidence, not qualification. Every benchmark must pin dataset manifest, artifact/processor versions, split keys, duplicate/near-duplicate checks, metric code, bootstrap intervals and failure accounting. The test set must remain sealed from calibration and threshold selection.

Role	Split design	Primary metrics	Required slices
Deepfake - facial	Identity/video/source/manipulation-separated; FaceForensics++ training relation recorded	AUROC/AUPRC, TPR at fixed low FPR, FPR, ECE/Brier, abstention coverage	local/partial edits, multiple/no face, alignment/occlusion, cross-dataset
Deepfake - synthetic	Generator family/version and source-image separated; generator-held-out final	class metrics + FPR on real cameras/art, selective risk, calibration	screenshot/transcode, resize, blur, JPEG, low resolution, unknown generators
Deepfake - forensic	No shared near-duplicate/source pipeline with semantic models	incremental lift and conditional error correlation, not standalone accuracy only	frequency-altering transforms, camera noise, recompression, OOD
Phishing - text	Campaign/time/sender/domain/tenant grouped	AUPRC, recall at FPR budgets, FPR, ECE/Brier, cost-weighted utility	BEC, financial fraud, smishing, legitimate urgency, newsletters, quoted history
Phishing - multilingual	Locale and campaign separated; code-switch/Hinglish declared	per-locale and macro metrics, worst-slice FPR, calibration	transliteration, mixed script, locale-specific lures, short SMS
Phishing - features	Message/entity/time grouping; auth/relationship features frozen as observed at event time	incremental lift, missingness robustness, calibration	SPF/DKIM/DMARC permutations, spoof/BEC, new senders, compromised accounts
URL multiclass	Domain/campaign/time grouped; test domains absent from train	per-class PR, confusion matrix, FPR/10k, calibration	new domains, compromised legitimate, shorteners, IDN, malformed/encoded, class ambiguity
URL feature/shadow	Same sealed split but separate feature pipeline	conditional error correlation and incremental detection at fixed FPR	PSL edge cases, special-use IP, confusables, very long URL
Page text	Page/campaign/domain/time grouped after safe collection	per-class PR, FPR on legitimate login/payment pages, calibration	cloaking, empty/text-light pages, multilingual, compromised brands
Brand embeddings	Tenant/reference split and unseen spoof templates	Recall@K, ROC for similarity evidence, false match rate	same-brand legitimate, subbrands, localization, crop/scale/theme changes

Benchmark acceptance is role-specific

- No universal accuracy threshold is proposed. Product harm is driven by false positives, missed attacks, evidence completeness and calibration under drift.
- Primary/shadow promotion requires a pre-registered success criterion and confidence intervals. Rerunning until a threshold wins is not acceptable.
- Provider/network failures, invalid outputs and abstentions count in end-to-end reliability; exclude none from operational availability metrics.
- India-relevant lures, English/Hinglish/code-switch and IDN/confusable slices must be explicitly represented, not inferred from 'multilingual' tags.

32. Calibration and threshold requirements

Raw model score is not an automatically calibrated probability. Softmax/sigmoid normalization only creates a distribution over the model's output classes; it does not establish empirical correctness for VeriTrust traffic.

Evidence type	Candidate method	Required evaluation	Version key
Binary neural classifier	Temperature scaling first; Platt/logistic when mapping one margin; isotonic only with enough data	ECE, Brier, reliability curve, NLL, slice calibration	model + processor + dataset window + calibrator ID
Multiclass URL	Vector/temperature scaling; classwise calibration if justified	classwise ECE/Brier, confusion and rare-class reliability	all four logits; never calibrate collapsed max alone
GBDT/linear	Platt/logistic or isotonic with held-out data	Brier/ECE and stability under missing features	feature schema + model + calibrator
Embedding similarity	Reference/tenant-specific empirical mapping or conformal/rank bands	false match rate, recall@K, coverage	encoder + processor + reference-index version
OOD	Distance/density threshold or conformal calibration on declared in/OOD sets	coverage, false-OOD/in-distribution rejection, selective risk	embedding + reference distribution + OOD calibrator
Gateway policy	Do not recalibrate by summing uncalibrated inputs	decision reliability by evidence-completeness stratum	policy version + upstream calibrated evidence IDs

Current hard-coded thresholds are provisional. Do not inherit 0.5, 0.45, 0.75, 0.90 or current weighted blends. Select thresholds from held-out cost/coverage objectives, freeze them with the calibrator, and monitor drift before retraining/recalibration.

33. Model independence and error-correlation strategy

Pair	Overlap analysis	Expected correlation	Recommendation
Community Forensics 384 + 224	Same architecture family/training data; resolution differs	Very high	Use 224 only as latency alternative/shadow, never independent evidence
Official Community Forensics + buildborderless integration	Intended same checkpoint family; integration/export differs	Very high	Reproducibility challenger only; prove equivalence
Community Forensics + yemandy	Full-frame generator vs aligned-face manipulation; small ViT vs CLIP-L; different data	Moderate/low by role	Good complementary roles; do not compare as interchangeable binary detectors
Semantic image detector + private SRM/DCT/residual	Semantic Transformer vs low-level forensic features	Expected lower	Preferred independent secondary; measure conditional correlation
PhishLens + dima/cybersectony DistilBERT	Same DistilBERT family, overlapping public corpora likely	High	Do not call independent; choose one fast primary
PhishLens + ModernBERT-large	Different architecture/context and calibration data, but text-only task overlap	Moderate	Useful shadow if campaign/time conditional errors differ
Text Transformer + private header/auth GBDT	Semantic text vs structured authentication/relationship facts	Expected lower	Preferred independent phishing evidence
URLBERT multiclass + URLBERT phishing	Same tiny BERT family and overlapping data	Very high	Binary model is correlated comparison only
URLBERT + pirocheto LinearSVM	Transformer token representation vs vectorizer/linear boundary	Lower expected	Useful public shadow; verify preprocessing overlap
URLBERT + private URL-feature GBDT	Token model vs PSL/IDN/lexical/relationship schema	Expected low	Preferred independent production secondary
URL model + page text + screenshot embedding	URL string vs rendered text vs visual reference	Low by modality	Complementary asynchronous evidence; collector bias still evaluated

Required statistic: report disagreement rate, error correlation on positives/negatives, conditional miss overlap, incremental recall at fixed FPR and gateway utility by evidence completeness. Repository-name diversity is never a substitute.

34. Recommended final VeriTrust model stack

Domain	Role	Preferred candidate	Class	Initial status
Deepfake	Face detector	OpenCV Zoo YuNet (non-HF)	D	Preprocessing only
Deepfake	Facial manipulation primary	yemandy/deepfake-detection	A	Disabled -> benchmark
Deepfake	Synthetic full-frame primary	OwensLab/commfor-model-384	A	Disabled -> benchmark
Deepfake	Synthetic latency shadow	OwensLab/commfor-model-224	A	Shadow only
Deepfake	Independent forensic	Private residual/SRM/DCT model	C	Train
Deepfake	Quality/OOD	Deterministic quality + private embedding OOD head	D/C	Build/train
Phishing	Fast classifier	Sonje03/phishlens-distilbert	A	Disabled -> benchmark
Phishing	Multilingual	microsoft/mdeberta-v3-base + VeriTrust head	B	Fine-tune
Phishing	Large shadow	mikaelnurminen/modernbert-large-phishing	A	Shadow after stability tests
Phishing	Structured features	Private VeriTrust GBDT	C	Train
Phishing	Semantic extractor	Mistral-7B-Instruct-v0.3	A	Optional / disabled
Link	Four-class URL primary	CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	A	Disabled -> benchmark
Link	Independent public shadow	pirocheto/phishing-url-detection ONNX	A	Shadow
Link	Static URL features	Private VeriTrust GBDT	C	Train
Link	Page text	microsoft/deberta-v3-small + VeriTrust head	B	Fine-tune
Link	Screenshot/brand	google/siglip2-base-patch16-224	A	Embedding evidence only
Gateway	Correlation	Deterministic versioned rules	D	No learned model initially
Learning	Core HF requirement	None	D	No model dependency

35. Final model acquisition queue - what to get from Hugging Face first

Priority	VeriTrust role	Exact HF repository	URL	Revision prefix	Params	Primary weight	Class	Action
1	Full-frame synthetic-image primary	OwensLab/commfor-model-384	HF	6076002bf0d9	21.8M	83.22 MiB	CLASS A	DOWNLOAD + SELF-HOST
2	Facial manipulation detector	yermandy/deepfake-detection	HF	9a6857ec642d	UNKNOWN	1.13 GiB	CLASS A	DOWNLOAD + SELF-HOST
3	Preferred fast English classifier candidate	Sonje03/phishlens-distilbert	HF	b585a2b9bbf6	67M	255.43 MiB	CLASS A	DOWNLOAD + SELF-HOST
4	Preferred fast four-class URL candidate	CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	HF	917ded0543a6	3.69M	14.08 MiB	CLASS A	DOWNLOAD + SELF-HOST
5	Preferred multilingual fine-tuning backbone	microsoft/mdeberta-v3-base	HF	a0484667b223	279M	1.24 GiB	CLASS B	FINE-TUNE FIRST
6	Large shadow/error-diversity classifier	mikaelnurminen/modernbert-large-phishing	HF	d28c16f88245	396M	755.01 MiB	CLASS A	BENCHMARK SHADOW
7	Independent linear-SVM phishing shadow	pirocheto/phishing-url-detection	HF	44f3b19f705b	UNKNOWN	22.46 MiB	CLASS A	DOWNLOAD ONNX + BENCHMARK
8	Sanitized page-text fine-tuning backbone	microsoft/deberta-v3-small	HF	a36c739020e0	142M	272.81 MiB	CLASS B	FINE-TUNE FIRST
9	Screenshot/approved-brand embedding model	google/siglip2-base-patch16-224	HF	75de2d55ec2d	375M	1.40 GiB	CLASS A	DOWNLOAD + SELF-HOST
10	Full-frame synthetic-image latency alternative	OwensLab/commfor-model-224	HF	26afc31e6b40	21.7M	82.66 MiB	CLASS A	KEEP SHADOW ONLY
11	New URL encoder + task-head challenger	CrabInHoney/urlbert-tiny-v5	HF	043ab17548c3	58.3M	222.33 MiB	CLASS A	BENCHMARK AFTER V4
12	Large English shadow classifier	ealvaradob/bert-finetuned-phishing	HF	fa8fb73a0071	336M	1.25 GiB	CLASS A	KEEP SHADOW ONLY
Non-HF/custom priority	Component	Action						
P0	OpenCV Zoo YuNet	Pin upstream commit and ONNX SHA-256; benchmark face recall/landmarks						
P0	Private residual/SRM/DCT forensic model	Design dataset/feature pipeline; train independent model						
P0	Private phishing feature GBDT	Define typed auth/sender/relationship schema; train						
P0	Private URL feature GBDT	Implement PSL/IDN/confusable-safe features; train						
P1	Private embedding OOD head	Define qualified distributions and calibrate abstention						

36. Environment-variable mapping and alias handoff

Current variable	Alias	Current expected contract	Recommended future model/component	Drop-in?	Required refactor
HF_DEEPFAKE_PIXEL_MODEL	Pixel	Raw-image binary HF label arrays	OwensLab/commfor-model-384 only after worker	No	DO NOT CONFIGURE UNTIL ADAPTER IMPLEMENTED; assign Pixel exclusively to full-frame synthesis or rename
HF_DEEPFAKE_PRISM_MODEL	Prism	Raw-image binary; silent Pixel fallback	yermandy/deepfake-detection only if Prism is redefined as face manipulation	No	Separate aligned-face worker; remove cross-role fallback; likely rename alias to communicate role
HF_PHISHING_MAILGUARD_MODEL	MailGuard	HF JSON {inputs}; guessed/collapsed labels	Sonje03/phishlens-distilbert	No	Explicit Safe/Phishing map, subject/body framing, self-host endpoint/worker
HF_PHISHING_CORTX_MODEL	Cortex	Featherless chat final verdict	Mistral-7B-Instruct-v0.3 optional	No	Provider currently not live; change prompt/output to bounded observations; disable truth/risk
HF_LINK_SWIFT_MODEL	Swift	HF JSON URL; current LABEL_1/3 map	CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	No	Correct map; self-host worker; preserve four classes; DO NOT CONFIGURE NOW
No variable	Sentinel	Locked/unavailable	Private URL feature GBDT	No	Add private registry/worker; rename Sentinel only after scientific contract is declared
HF_MODEL_PATHS	Alias map	Mutable JSON owner/repo IDs	No direct model IDs recommended	No	Replace with immutable registry keys/bundles; resolve before inference
HF_MODEL_PATH	Legacy Pixel	Legacy Pixel fallback	None	No	Deprecate
HF_TOKEN / HF_ACCESS_TOKEN	Provider auth	Bearer/provider auth	Scoped provider credentials	N/A	Secret management/data-flow review; not artifact identity

No recommended target model ID should be placed directly into the current variables during this research phase. The only investigated configuration-only image compatibility fixture is `dima806/deepfake_vs_real_image_detection`; its author warns of concept drift, so it is not the target production model.

37. Implementation handoff to Codex

The implementation session should treat each entry below as an immutable internal scientific key, not a UI alias. It must build registry/worker/evidence/test scaffolding before enabling any provider or model.

Implementation handoff 1/16

deepfake.face_preprocess.yunet - Face detector/landmarks

Internal model key	deepfake.face_preprocess.yunet
Role	Face detector/landmarks
Exact repository/revision	OpenCV Zoo YuNet
Required artifacts	TBD - private training/component acquisition must create SHA-256 and SBOM
Tokenizer/processor	Defined by component/training contract
Runtime	OpenCV DNN / ONNX CPU
Adapter type	Preprocessing worker
Input contract	Canonical decoded image
Output contract	boxes, 5 landmarks, detection score, aligned-crop child IDs
Label map	No authenticity label
Expected config	Private registry component with upstream commit + ONNX SHA-256
Current source/new service	new deepfake preprocessing worker; gateway child-artifact manifest
Tests before activation	decode/orientation, multi/no-face, landmark/alignment golden fixtures
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 2/16

deepfake.face_manipulation.clip_l14_In - Facial manipulation primary

Internal model key	deepfake.face_manipulation.clip_l14_In
Role	Facial manipulation primary
Exact repository/revision	vermandy/deepfake-detection@9a6857ec642deb57373c5437be803a199468b8c6
Required artifacts	model.ckpt (1.13 GiB)
Tokenizer/processor	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files
Runtime	Self-hosted GPU reviewed project runtime
Adapter type	Aligned-face binary worker
Input contract	Pinned aligned face artifact
Output contract	raw logit + normalized real/manipulated scores
Label map	Verify sign/order from pinned upstream inference + fixtures
Expected config	DO NOT CONFIGURE UNTIL ADAPTER IMPLEMENTED
Current source/new service	new deepfake face worker; model registry; evidence schema
Tests before activation	DeepfakeBench preprocessing parity, real/manipulated golden faces, cross-dataset
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 3/16

deepfake.synthetic.commmfor384 - Full-frame synthetic primary

Internal model key	deepfake.synthetic.commmfor384
Role	Full-frame synthetic primary
Exact repository/revision	OwensLab/commfor-model-384@6076002bf0d9dd37537f965ee2f06f826c333b61
Required artifacts	model.safetensors (83.22 MiB)
Tokenizer/processor	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files
Runtime	Self-hosted PyTorch GPU; reviewed mixin
Adapter type	Full-frame synthetic worker
Input contract	Pinned 384 preprocessing over full-frame child
Output contract	raw binary score + calibrated generated probability
Label map	real=0, fake=1 convention; wrapper output fixture required
Expected config	DO NOT CONFIGURE UNTIL ADAPTER IMPLEMENTED
Current source/new service	new synthetic-image worker; registry; orchestrator evidence mapping
Tests before activation	official notebook parity, generator-held-out, screenshot/compression/OOD
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 4/16

deepfake.synthetic.commfor224_shadow - Full-frame latency shadow

Internal model key	deepfake.synthetic.commfor224_shadow
Role	Full-frame latency shadow
Exact repository/revision	OwensLab/commfor-model-224@26afc31e6b40c312c3fd42c05a758be62446215b
Required artifacts	model.safetensors (82.66 MiB)
Tokenizer/processor	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files
Runtime	Self-hosted PyTorch CPU/GPU
Adapter type	Full-frame synthetic shadow worker
Input contract	Pinned 224 preprocessing
Output contract	raw binary score + separate calibrator
Label map	real=0, fake=1 convention; fixture required
Expected config	Separate shadow key - never Prism/Pixel silent fallback
Current source/new service	same worker family; independent version/calibrator entry
Tests before activation	384-vs-224 latency/quality/correlation and parity
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	shadow

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 5/16

deepfake.forensic.residual_v1 - Independent residual/frequency forensic

Internal model key	deepfake.forensic.residual_v1
Role	Independent residual/frequency forensic
Exact repository/revision	CUSTOM VERITRUST MODEL
Required artifacts	TBD - private training/component acquisition must create SHA-256 and SBOM
Tokenizer/processor	Defined by component/training contract
Runtime	Self-hosted ONNX/CPU or modest GPU
Adapter type	Residual/SRM/DCT worker
Input contract	Versioned residual/frequency child tensor
Output contract	raw forensic score + calibrated evidence + abstain
Label map	real-like vs manipulation/generation artifact; define during training
Expected config	Private artifact registry only
Current source/new service	new forensic feature generator + model worker + calibrator
Tests before activation	leakage-safe generator/source split, transform robustness, independence
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 6/16

deepfake.ood.embedding_v1 - Image/face OOD evidence

Internal model key	deepfake.ood.embedding_v1
Role	Image/face OOD evidence
Exact repository/revision	CUSTOM VERITRUST OOD HEAD
Required artifacts	TBD - private training/component acquisition must create SHA-256 and SBOM
Tokenizer/processor	Defined by component/training contract
Runtime	Self-hosted CPU/GPU
Adapter type	Embedding distance/density/conformal worker
Input contract	Pinned embedding + quality metadata
Output contract	OOD score, in/out/unknown band, coverage
Label map	Qualified-distribution membership only
Expected config	Private reference distribution + calibrator versions
Current source/new service	OOD worker; registry distribution artifacts; abstention policy
Tests before activation	held-out generators/sources, near-OOD, calibration/coverage
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 7/16

phishing.text.fast.phishlens - Fast English phishing primary

Internal model key	phishing.text.fast.phishlens
Role	Fast English phishing primary
Exact repository/revision	Sonje03/phishlens-distilbert@b585a2b9bbf6777b344d017f4eaff9423363a489
Required artifacts	model.safetensors (255.43 MiB)
Tokenizer/processor	tokenizer_class=BertTokenizer; model_max_length=512
Runtime	Self-hosted Transformers/ONNX CPU
Adapter type	Explicit two-class text worker
Input contract	Pinned subject/body framing; max 256 baseline
Output contract	two logits + calibrated phishing probability
Label map	0 Safe; 1 Phishing
Expected config	DO NOT use MailGuard until safe/phishing map is explicit
Current source/new service	phishing worker/adaptor; config registry; evidence normalizer
Tests before activation	tokenizer parity, campaign/time/sender/locale slices, long-message aggregation
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 8/16

phishing.text.multilingual.mdeberta_v1 - Multilingual phishing classifier

Internal model key	phishing.text.multilingual.mdeberta_v1
Role	Multilingual phishing classifier
Exact repository/revision	microsoft/mdeberta-v3-base@a0484667b22365f84929a935b5e50a51f71f159d
Required artifacts	pytorch_model.bin (1.24 GiB); pytorch_model.generator.bin (905.32 MiB)
Tokenizer/processor	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files
Runtime	Fine-tune then self-host Transformers/ONNX
Adapter type	Locale-aware text worker
Input contract	Training-defined subject/body framing and tokenizer
Output contract	binary/multiclass logits + locale/calibration metadata
Label map	To be defined by VeriTrust dataset; base has no head
Expected config	FINE-TUNE FIRST
Current source/new service	training pipeline; dataset ledger; multilingual worker/calibrators
Tests before activation	English/Hinglish/declared locales, code-switch, campaign/time split
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 9/16

phishing.text.shadow.modernbert - Large phishing shadow

Internal model key	phishing.text.shadow.modernbert
Role	Large phishing shadow
Exact repository/revision	mikaelnurminen/modernbert-large-phishing@d28c16f882454115d71375d3321fbfd90e566490
Required artifacts	model.safetensors (755.01 MiB)
Tokenizer/processor	tokenizer_class=PreTrainedTokenizer; model_max_length=8192
Runtime	Self-hosted GPU; eager attention; batch=1 initially
Adapter type	Long-text shadow worker
Input contract	Pinned tokenizer; max 2048 baseline pending benchmark
Output contract	two logits + separate calibrator
Label map	0 benign; 1 phishing
Expected config	Shadow-only registry key
Current source/new service	shadow worker; no user-facing direct alias
Tests before activation	NaN/Inf, variable length, quoted threads, real-world holdout
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	shadow

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 10/16

phishing.features.gbdt_v1 - Structured phishing feature model

Internal model key	phishing.features.gbdt_v1
Role	Structured phishing feature model
Exact repository/revision	CUSTOM VERITRUST GBDT
Required artifacts	TBD - private training/component acquisition must create SHA-256 and SBOM
Tokenizer/processor	Defined by component/training contract
Runtime	LightGBM/XGBoost exported to ONNX CPU
Adapter type	Typed feature-vector worker
Input contract	Auth/sender/domain/relationship/content feature schema
Output contract	raw margin + calibrated phishing evidence + missingness
Label map	Decision-quality phishing label with campaign/time split
Expected config	Private artifact + feature-schema version
Current source/new service	feature extractor, GBDT worker, feature/evidence registry
Tests before activation	missing headers, spoofing/BEC, temporal drift, SHAP sanity, calibration
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 11/16

phishing.extractor.mistral7b_v03 - Optional bounded observation extractor

Internal model key	phishing.extractor.mistral7b_v03
Role	Optional bounded observation extractor
Exact repository/revision	mistralai/Mistral-7B-Instruct-v0.3@c170c708c41dac9275d15a8fff4eca08d52bab71
Required artifacts	consolidated.safetensors (13.50 GiB); model-00001-of-00003.safetensors (4.61 GiB); model-00002-of-00003.safetensors (4.66 GiB); model-00003-of-00003.safetensors (4.23 GiB)
Tokenizer/processor	tokenizer_class=LlamaTokenizer; model_max_length=100000000000000019884624838656
Runtime	Self-hosted GPU only if approved
Adapter type	Schema-constrained extraction service
Input contract	Minimized message text; no privileged headers/secrets
Output contract	credential/payment/urgency/impersonation observations + spans
Label map	No phishing truth/risk/action
Expected config	Disabled; no current live provider
Current source/new service	separate privacy-reviewed extractor; schema validator
Tests before activation	prompt injection, JSON conformance, evidence-span fidelity, PII/privacy
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 12/16

link.url.multiclass.urlbert_v4 - Fast four-class URL primary

Internal model key	link.url.multiclass.urlbert_v4
Role	Fast four-class URL primary
Exact repository/revision	CrabInHoney/urlbert-tiny-v4-malicious-url-classifier@917ded0543a630d6e82570bb01ae692f6cbb95f1
Required artifacts	model.safetensors (14.08 MiB)
Tokenizer/processor	tokenizer_class=BertTokenizer; model_max_length=64
Runtime	Self-hosted Transformers/ONNX CPU
Adapter type	Explicit four-class URL worker
Input contract	Recorded raw/model URL string; max 64 tokens
Output contract	four logits/probabilities + calibrated class evidence
Label map	0 benign; 1 defacement; 2 malware; 3 phishing
Expected config	DO NOT set HF_LINK_SWIFT_MODEL before map/worker fix
Current source/new service	link worker; correct normalizer; registry; evidence schema
Tests before activation	tokenizer/ONNX parity, IDN/encoding/long URL, temporal/domain splits
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 13/16

link.url.shadow.linear_svm - Independent lexical phishing shadow

Internal model key	link.url.shadow.linear_svm
Role	Independent lexical phishing shadow
Exact repository/revision	pirocheto/phishing-url-detection@44f3b19f705b52532e0aadf3d0d15dd892b8a2fb
Required artifacts	model.onnx (22.46 MiB)
Tokenizer/processor	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files
Runtime	ONNX Runtime CPU
Adapter type	ONNX string-input worker
Input contract	Recorded URL string tensor
Output contract	class + probability outputs
Label map	index 0 benign; index 1 phishing
Expected config	Acquire model.onnx only; reject model.pkl
Current source/new service	ONNX link-shadow worker; model registry
Tests before activation	output nodes/types, parity, IDN/malformed URLs, independence
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	shadow

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 14/16

link.features.gbdt_v1 - Static URL feature primary secondary

Internal model key	link.features.gbdt_v1
Role	Static URL feature primary secondary
Exact repository/revision	CUSTOM VERITRUST GBDT
Required artifacts	TBD - private training/component acquisition must create SHA-256 and SBOM
Tokenizer/processor	Defined by component/training contract
Runtime	ONNX Runtime CPU
Adapter type	Typed URL-feature worker
Input contract	PSL/IDN/confusable/lexical/encoding/host/path/query/relationship features
Output contract	multiclass/binary margin + calibrated evidence + missingness
Label map	Define from campaign/time-qualified data
Expected config	Private artifact + PSL/feature-schema versions
Current source/new service	URL parser/features; PSL updater; GBDT worker/registry
Tests before activation	PSL changes, special-use addresses, IDN/confusable, temporal/domain holdouts
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 15/16

link.page_text.deberta_small_v1 - Sanitized page-text classifier

Internal model key	link.page_text.deberta_small_v1
Role	Sanitized page-text classifier
Exact repository/revision	microsoft/deberta-v3-small@sha384-a36c739020e01763fe789b4b85e2df55d6180012
Required artifacts	pytorch_model.bin (272.81 MiB)
Tokenizer/processor	Repository-specific upstream preprocessing; not fully encoded in standard Hub processor files
Runtime	Fine-tune then self-host ONNX/CPU-GPU
Adapter type	Async sanitized-text worker
Input contract	Sandbox title + visible text + form labels only
Output contract	task logits + calibrated page evidence
Label map	Define during VeriTrust fine-tune
Expected config	FINE-TUNE FIRST
Current source/new service	sandbox output schema; async page-text worker; registry
Tests before activation	no-script/no-secret contract, chunking, compromised legitimate pages, campaign split
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

Implementation handoff 16/16

link.brand.siglip2_base - Screenshot/brand embedding evidence

Internal model key	link.brand.siglip2_base
Role	Screenshot/brand embedding evidence
Exact repository/revision	google/siglip2-base-patch16-224@75de2d55ec2d0b4efc50b3e9ad70dba96a7b2fa2
Required artifacts	model.safetensors (1.40 GiB)
Tokenizer/processor	size={'height': 224, 'width': 224}; do_resize=True; do_normalize=True; do_rescale=True; rescale_factor=0.00392156862745098; image_mean=[0.5, 0.5, 0.5]; image_std=[0.5, 0.5, 0.5]; resample=2; image_processor_type=SiglipImageProcessor
Runtime	Self-hosted Transformers GPU
Adapter type	Embedding + approved-reference index worker
Input contract	Pinned 224 RGB screenshot + tenant reference set
Output contract	embedding, nearest references, similarity/raw rank
Label map	No malicious label
Expected config	Separate embedding/reference-index versions
Current source/new service	screenshot embedding worker; approved-brand index; evidence schema
Tests before activation	processor parity, crop/scale, benign brand pages, spoof variants, tenant isolation
Benchmark/calibration	Role benchmark in section 31 + dedicated versioned calibration in section 32
Initial deployment status	disabled

Activation sequence: disabled -> local reproduction/golden fixtures (R3) -> benchmark/calibration/security/legal gates -> shadow (R4) -> canary candidate (R5). Public information alone does not cross R2.

38. Models and components VeriTrust must train itself

Required item	Class	Training input	Output	Core qualification
Private residual/frequency forensic detector	C	SRM/DCT/noise/residual child artifacts	Independent forensic evidence	Generator/source/transform separated; parity across preprocessing
Private embedding OOD head	C	Pinned image/face embeddings + quality	OOD distance/coverage/abstain	Declared in-distribution and near/far-OOD calibration
mDeBERTa multilingual phishing head	B	Locale-aware subject/body text	Phishing evidence per locale	Campaign/time/locale/code-switch split
Private phishing structured GBDT	C	Auth/sender/domain/relationship/content features	Independent phishing score	Message/entity/time split; missingness and drift
Private URL structured GBDT	C	PSL/IDN/confusable/lexical/network relationship features	Binary/multiclass URL evidence	Domain/campaign/time split; feature/PSL versions
DeBERTa sanitized page-text head	B	Sandbox title/visible text/form labels	Page phishing/impersonation evidence	Domain/campaign/time split; no-script contract
CLIP facial head alternative	B	Aligned faces	Optional VeriTrust face manipulation head	Only if yermandy reproduction/benchmark is inadequate
XLM-R multilingual fallback head	B	Locale-aware text	Alternative multilingual evidence	Only if it beats mDeBERTa on pre-registered criteria

Count used for the final task handoff: 8 acquisition/training items require VeriTrust fine-tuning or custom training (four Class B backbones and four mandatory private Class C model families). Optional fallback fine-tunes are research choices, not simultaneous requirements.

39. Non-Hugging-Face dependencies and connectors

Component	Role	Acquisition/implementation note
OpenCV Zoo YuNet	Face boxes + landmarks	Pin upstream commit/model hash; OpenCV DNN runtime; preprocessing only
Public Suffix List	Registrable-domain interpretation	Pin snapshot/version; test updates and private suffixes
IDNA/confusables	Unicode/IDN evidence	Use standards-based library/tables; version confusable data
LightGBM/XGBoost/ONNX Runtime	Private structured models	Lock dependencies; export/validate ONNX; record feature schema
Google Web Risk	Reputation connector	External fact source; latency/quota/privacy/licensing; not a model
VirusTotal	Multi-engine reputation	Connector evidence with terms/retention controls; not ground truth
URLhaus / PhishTank	Threat-feed evidence	Timestamp/source/TTL; false-positive and licensing review
DNS/RDAP/TLS/HTTP	Network/domain observations	Async collectors; SSRF controls; typed partial failures
Browser/sandbox	Sanitized page text + screenshot	Isolated egress, no secrets, no script trust, content-size/time limits
Supabase	Registry/evidence/benchmark/calibration persistence	RLS, immutable artifact receipts, retention and tenant separation

40. Rejected and unavailable model appendix

Repository/component	Intended role	Status	Rejection reason	License gate
onnx-community/CommunityForensics-DeepfakeDet-ViT-ONNX	Community Forensics ONNX conversion	REJECTED	Conversion predates the July 2026 corrected integration and lacks an authoritative equivalence chain. Use a fresh export from a pinned verified checkpoint instead.	REJECT
prithviMLmods/AI-vs-Deepfake-vs-Real	Three-way image classifier	REJECTED	The card itself says it is not a deepfake detector; its Artificial/Deepfake/Real taxonomy merges distinct VeriTrust roles and current code would normalize it incorrectly.	REJECT PENDING DATA PROVENANCE
aamoshdahal/email-phishing-distilbert-finetuned	Binary DistilBERT candidate	REJECTED	Task fit is plausible, but the Hub repository declares no license and config labels are generic. Do not acquire for product use until the owner supplies explicit licensing and a pinned label contract.	LICENSE UNKNOWN - REJECT
JustAPR/email-phishing-distilbert-finetuned	Copied binary DistilBERT card	REJECTED	Repository owner differs from the model ID throughout the copied card; config lacks a verified label map and no license is declared. Supply-chain/provenance failure.	LICENSE UNKNOWN - REJECT
Qwen/Qwen2.5-3B-Instruct	Optional bounded extractor alternative	REJECTED	Technically attractive 3.09B JSON-capable model with a live Featherless provider, but its Qwen Research License permits non-commercial use only unless a separate commercial license is obtained.	NON-COMMERCIAL - REJECT
elftsdmr/malware-url-detect	Turkey-focused URL classifier	REJECTED	The card states an unknown dataset, Turkey-specific harmful links and domain-only input. This is not a qualified multiclass URL model and has weak provenance.	REJECT PENDING DATA PROVENANCE
protectai/codebert-base-Malicious_URLs-onnx	CodeBERT ONNX malicious URL model	VERIFIED - ARCHIVED/STALE	The card explicitly says the project is archived and unmaintained; model license is absent. A fresh, licensed export from an active source would be required.	LICENSE UNKNOWN - REJECT
DiligentAI/urlbert-url-classifier	Official-site vs platform URL classifier	REJECTED	It classifies official websites versus third-party platforms, not benign versus malicious. Repository name similarity is not task fit.	REJECT
Generic 'gateway AI model'	Unified gateway	REJECTED	Handbooks require deterministic versioned correlation initially; no acquisition role exists.	N/A
Generic Learning Module LLM	Learning core	REJECTED	Learning architecture does not require HF inference; adding one would manufacture a dependency.	N/A
VeriTrust Sentinel as an unnamed HF model	Link secondary	NOT FOUND / REJECTED	Sentinel is a locked UI alias with no configured model path; do not invent a repository.	N/A

41. Remaining research questions

- Can the official Community Forensics Hub mixin be reproduced byte/logit-equivalently in a minimal reviewed worker, and does the third-party corrected integration match it?
- What exact DeepfakeBench commit/config, face alignment and frame aggregation are required for the pinned yermandy revision?
- Which VeriTrust traffic locales, channel mix and PII/data-residency constraints determine the multilingual phishing corpus and whether any external LLM provider is permissible?
- Can public phishing datasets be legally and scientifically combined without campaign/sender duplicates or temporal leakage?
- What decision-labeling and appeal workflow will produce trustworthy structured-model and future gateway training labels?
- Which URLBERT input form (raw vs current normalized URL) reproduces card behavior and maximizes temporal/domain-held-out performance?
- Does URLBERT-v5's four-class head materially outperform v4 under VeriTrust splits enough to justify custom code and 16x parameters?
- What approved-brand reference governance, tenant isolation and refresh cadence are required for SigLIP2 evidence?
- What provider/regional/privacy contracts are acceptable for any managed endpoint receiving emails, URLs or screenshots?
- What operational FPR budgets and review capacity determine calibrated thresholds and gateway action policies?

42. Final implementation readiness checklist

Level	Meaning	Required gate	Current assessment
R0 Discovered	Existence only	Hub/API record captured	Most rejected/optional candidates
R1 Acquisition ready	Repository, SHA, artifacts, license identified	Pinned download manifest + legal-review ticket	Official Community Forensics, yermandy, bases, Mistral
R2 Reproduction ready	Processor/runtime/output contract understood	Local deterministic inference and explicit labels	PhishLens, ModernBERT, URLBERT-v4, pirocheto, SigLIP2 candidates
R3 Benchmark ready	Golden fixtures and parity pass	Reproducible worker/container + benchmark manifest	None from public evidence alone
R4 Shadow ready	Benchmark/calibration/security/legal gates pass	Shadow evidence with no user action	None yet
R5 Canary candidate	All verification except production rollout	Canary plan, SLOs, rollback, monitoring	None yet

Implementation start checklist

- Create immutable registry schema and resolve model/processor/calibrator before inference.
- Implement explicit per-role adapters/workers; delete label guessing and scientific cross-role fallback.
- Acquire only pinned artifact allowlists; hash, scan, mirror, archive licenses/cards and create SBOMs.
- Reproduce each R1/R2 candidate with golden tensors/fixtures and explicit invalid-output behavior.
- Build leakage-safe benchmark manifests and pre-register decision metrics/slices.
- Fit/version calibrators; preserve raw, normalized and calibrated fields.
- Implement expected-evidence ledger, typed abstention/unavailable states and deterministic gateway policy.
- Complete privacy/security/legal reviews before provider or shadow data flows.
- Advance disabled -> shadow -> canary only through readiness gates; no public card implies R5.

Final recommendation table - implementation planning authority

VeriTrust role	Final preferred candidate	HF link	Params	Download/repo size	Deployment	Current code compatibility	Readiness	Next action
Face detector/landmarks	OpenCV Zoo YuNet	N/A	TBD	TBD	OpenCV DNN / ONNX CPU	New component	R0	BUILD/TRAIN
Facial manipulation primary	yemandy/deepfake-detection	HF	UNKNOWN	1.70 GiB	Self-hosted GPU reviewed project runtime	Preprocessed aligned-face worker only	R1	DOWNLOAD + SELF-HOST
Full-frame synthetic primary	OwensLab/commfor-model-384	HF	21.8M	83.22 MiB	Self-hosted PyTorch GPU; reviewed mixin	Dedicated worker only	R1	DOWNLOAD + SELF-HOST
Full-frame latency shadow	OwensLab/commfor-model-224	HF	21.7M	82.66 MiB	Self-hosted PyTorch CPU/GPU	Dedicated worker only	R1	KEEP SHADOW ONLY
Independent residual/frequency forensic	CUSTOM VERITRUST MODEL	N/A	TBD	TBD	Self-hosted ONNX/CPU or modest GPU	New component	R0	BUILD/TRAIN
Image/face OOD evidence	CUSTOM VERITRUST OOD HEAD	N/A	TBD	TBD	Self-hosted CPU/GPU	New component	R0	BUILD/TRAIN
Fast English phishing primary	Sonje03/phishlens-distilbert	HF	67M	255.43 MiB	Self-hosted Transformers/ONNX CPU	Label-map and self-host worker required	R2	DOWNLOAD + SELF-HOST
Multilingual phishing classifier	microsoft/mdeberta-v3-base	HF	279M	5.45 GiB	Fine-tune then self-host Transformers/ONNX	No classification head; new worker	R1	FINE-TUNE FIRST
Large phishing shadow	mikaelnurminen/modernbert-large-phishing	HF	396M	4.20 GiB	Self-hosted GPU; eager attention; batch=1 initially	Dedicated worker; never current adapter	R2	BENCHMARK SHADOW
Structured phishing feature model	CUSTOM VERITRUST GBDT	N/A	TBD	TBD	LightGBM/XGBoost exported to ONNX CPU	New component	R0	BUILD/TRAIN
Optional bounded observation extractor	mistralai/Mistral-7B-Instruct-v0.3	HF	7.25B	27.00 GiB	Self-hosted GPU only if approved	Chat adapter shape fits, but provider is error and prompt role is wrong	R1	KEEP DISABLED / SELF-HOST LAB
Fast four-class URL primary	CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	HF	3.69M	14.08 MiB	Self-hosted Transformers/ONNX CPU	Label-map swap and self-host worker required	R2	DOWNLOAD + SELF-HOST
Independent lexical phishing shadow	pirocheto/phishing-url-detection	HF	UNKNOWN	1.49 GiB	ONNX Runtime CPU	Dedicated ONNX adapter	R2	DOWNLOAD ONNX + BENCHMARK
Static URL feature primary secondary	CUSTOM VERITRUST GBDT	N/A	TBD	TBD	ONNX Runtime CPU	New component	R0	BUILD/TRAIN
Sanitized page-text classifier	microsoft/deberta-v3-small	HF	142M	3.59 GiB	Fine-tune then self-host ONNX/CPU-GPU	No task head; async worker required	R1	FINE-TUNE FIRST
Screenshot/brand embedding evidence	google/siglip2-base-patch16-224	HF	375M	2.83 GiB	Self-hosted Transformers GPU	Embedding worker and reference index required	R2	DOWNLOAD + SELF-HOST

This table is sufficient to start implementation planning, but activation still requires every gate in sections 31, 32, 30 and 42.

43. References and direct Hugging Face links

Hugging Face repositories - exact revisions are recorded in the individual dossiers. All were accessible and examined on the research date.

Repository	Direct URL	Research revision
OwensLab/commfor-model-384	https://huggingface.co/OwensLab/commfor-model-384	6076002bf0d9dd3753f965ee2f06f826c333b61
OwensLab/commfor-model-224	https://huggingface.co/OwensLab/commfor-model-224	26afc31e6b40c312c3fd42c05a758be62446215b
yermandy/deepfake-detection	https://huggingface.co/yermandy/deepfake-detection	9a6857ec642deb57373c5437be803a199468b8c6
openai/clip-vit-large-patch14	https://huggingface.co/openai/clip-vit-large-patch14	32bd64288804d66eefd0ccbe215aa642df71cc41
buildborderless/CommunityForensics-DeepfakeDet-VIT	https://huggingface.co/buildborderless/CommunityForensics-DeepfakeDet-VIT	ac6ee457bea904a373065754107451793b56db00
onnx-community/CommunityForensics-DeepfakeDet-VIT-ONNX	https://huggingface.co/onnx-community/CommunityForensics-DeepfakeDet-VIT-ONNX	f9fc6516b37cd7d1bec3b94847b3682ce97249f0
dima806/deepfake_vs_real_image_detection	https://huggingface.co/dima806/deepfake_vs_real_image_detection	29e4cf9efc543845610045f6ba7e88e5cf9d9301
capcheck/ai-image-detection	https://huggingface.co/capcheck/ai-image-detection	a6661e07d38f1a097bba07ca9415538819278f09
prithivMLmods/Al-vs-Deepfake-vs-Real	https://huggingface.co/prithivMLmods/Al-vs-Deepfake-vs-Real	e1df5230c067ff4b189230343e548c365d80bbdb
cybersectony/phishing-email-detection-distilbert_v2.4.1	https://huggingface.co/cybersectony/phishing-email-detection-distilbert_v2.4.1	6724dec17360ab1f6450553f1d950404f6a5dde
dima806/phishing-email-detection	https://huggingface.co/dima806/phishing-email-detection	310a5602280486e01dd60e37230980cfc327b41
aamoshdahal/email-phishing-distilbert-finetuned	https://huggingface.co/aamoshdahal/email-phishing-distilbert-finetuned	f9d211125fcb6e85cf73630957667bc2a6e93ed3
ealvaradob/bert-finetuned-phishing	https://huggingface.co/ealvaradob/bert-finetuned-phishing	fa8fb73a007174c410ab7160d4e4c6e6b8d998d4
microsoft/mdeberta-v3-base	https://huggingface.co/microsoft/mdeberta-v3-base	a0484667b22365f84929a935b5e50a51f71f159d
mistralai/Mistral-7B-Instruct-v0.3	https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3	c170c708c41dac9275d15a8ff4eca08d52bab71
FacebookAI/xlm-roberta-base	https://huggingface.co/FacebookAI/xlm-roberta-base	e73636d4f797dec63c3081bb6ed5c7b0bb3f2089
Sonje03/phishtens-distilbert	https://huggingface.co/Sonje03/phishtens-distilbert	b585a2b9bbf6777b344d017f4eaff9423363a489
mikaelnurminen/modernbert-large-phishing	https://huggingface.co/mikaelnurminen/modernbert-large-phishing	d28c16f882454115d71375d3321fbfd90e566490
JustAPR/email-phishing-distilbert-finetuned	https://huggingface.co/JustAPR/email-phishing-distilbert-finetuned	9d8e0ac6797de33a3c97e2fc13fe8f99b1b12dfc
Qwen/Qwen2.5-3B-Instruct	https://huggingface.co/Qwen/Qwen2.5-3B-Instruct	aa8e72537993ba99e69dffa59ed015b17504d1
CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	https://huggingface.co/CrabInHoney/urlbert-tiny-v4-malicious-url-classifier	917ded0543a630d6e82570bb01ae692f6cbb95f1
CrabInHoney/urlbert-tiny-v4-phishing-classifier	https://huggingface.co/CrabInHoney/urlbert-tiny-v4-phishing-classifier	fd962a5cd04e20ceed46aa8a6e31a2b40d4c8916
CrabInHoney/urlbert-tiny-v5	https://huggingface.co/CrabInHoney/urlbert-tiny-v5	043ab17548c371ef8a0ce6d3f7c728e48f844e7d
elftsdmr/malware-url-detect	https://huggingface.co/elftsdmr/malware-url-detect	647f1adf5305e90235ce2a741b9c3351c22ae458
microsoft/deberta-v3-small	https://huggingface.co/microsoft/deberta-v3-small	a36c739020e01763fe789b4b85e2df55d6180012
google/siglip2-base-patch16-224	https://huggingface.co/google/siglip2-base-patch16-224	75de2d55ec2d0b4efc50b3e9ad70dba96a7b2fa2
protectai/codebert-base-Malicious_URLs-onnx	https://huggingface.co/protectai/codebert-base-Malicious_URLs-onnx	6e77d4e5b602736f8ec048463242e41894884844
pirocheto/phishing-url-detection	https://huggingface.co/pirocheto/phishing-url-detection	44f3b19f705b52532e0aadf3d015dd892b8a2fb
DiligentAI/urlbert-url-classifier	https://huggingface.co/DiligentAI/urlbert-url-classifier	e7b707843407d45dfcfc2844517eaf5ef56a63c

Authoritative upstream and platform sources

Source	Direct URL
Hugging Face Hub API provider mapping	https://huggingface.co/docs/inference-providers/hub-api
Hugging Face Inference Providers	https://huggingface.co/docs/inference-providers/index
Hugging Face Inference Endpoints	https://huggingface.co/docs/inference-endpoints/index
Hugging Face pickle security	https://huggingface.co/docs/hub/security-pickle
Safetensors documentation	https://huggingface.co/docs/safetensors/index
Community Forensics official code	https://github.com/JeongsooP/Community-Forensics
Community Forensics paper	https://arxiv.org/abs/2411.04125
Yermakov et al. official code	https://github.com/vermandy/deepfake-detection
Yermakov et al. paper	https://arxiv.org/abs/2503.19683
OpenCV Zoo YuNet	https://github.com/opencv/opencv_zoo/tree/main/models/face_detection_yunet
Public Suffix List	https://publicsuffix.org/
Google Web Risk	https://cloud.google.com/security/products/web-risk
VirusTotal API	https://docs.virustotal.com/reference/overview
URLhaus API	https://urlhaus-api.abuse.ch/
PhishTank	https://phishtank.org/
ONNX Runtime	https://onnxruntime.ai/docs/
ReportLab	https://www.reportlab.com/documentation/

Repository-local architecture sources

Document	Path	Pages
Architecture blueprint	inputs/pdfs/VeriTrust_Four_Core_Detection_Architecture_Blueprint_2026-08-09-1.pdf	22
Deepfake implementation handbook	output/pdfs/VeriTrust_Deepfake_Detection_Implementation_Handbook.pdf	26
Phishing implementation handbook	output/pdfs/VeriTrust_Phishing_Detection_Implementation_Handbook.pdf	25
Learning Module implementation handbook	output/pdfs/VeriTrust_Learning_Module_Implementation_Handbook.pdf	22
Link implementation handbook	output/pdfs/VeriTrust_Link_Detection_Implementation_Handbook.pdf	27
Unified Gateway implementation handbook	output/pdfs/VeriTrust_Unified_Gateway_Implementation_Handbook.pdf	34

End of report. Point-in-time research verified on 15 August 2026. Re-run Hub metadata/provider/license checks immediately before acquisition or implementation.